CS177 Bioinformatics: Software Development and Applications

Lecture 22: Synthetic Biology and Protein Design

Jie Zheng (郑杰)
PhD, Associate Professor
School of Information Science and Technology (SIST), ShanghaiTech University

Spring, 2024

Outline

- Overview of synthetic biology
  - Engineering DNA
  - Biomolecular engineering
  - Host engineering (to skip here)
  - Data science
- AI for protein design
  - ProGen and ProGen2
  - RFdiffusion

Overview of Synthetic Biology

Slides adapted from EBRC (Engineering Biology Research Consortium):
https://ebrc.org/wp-content/uploads/2022/09/Intro-to-Synthetic-Biology-1.pdf

What is Synthetic Biology?

Engineering Biology (a broader term for Synthetic Biology) is the **design and construction of new biological entities** such as enzymes, genetic circuits, and cells or the redesign of existing biological systems.

Engineering biology **builds on the advances** in molecular, cell, and systems biology and **seeks to transform** biology in the same way that synthesis transformed chemistry and integrated circuit design transformed computing.

The element that distinguishes engineering biology from traditional molecular and cellular biology is the **focus on the design and construction of core components** (e.g., parts of enzymes, genetic circuits, metabolic pathways) that can be modeled, understood, and tuned to meet specific performance criteria, and the assembly of these smaller parts and devices into larger integrated systems to solve specific problems.

Unlike many other areas of engineering, biology is incredibly non-linear and less predictable, and there is less knowledge of the parts and how they interact.

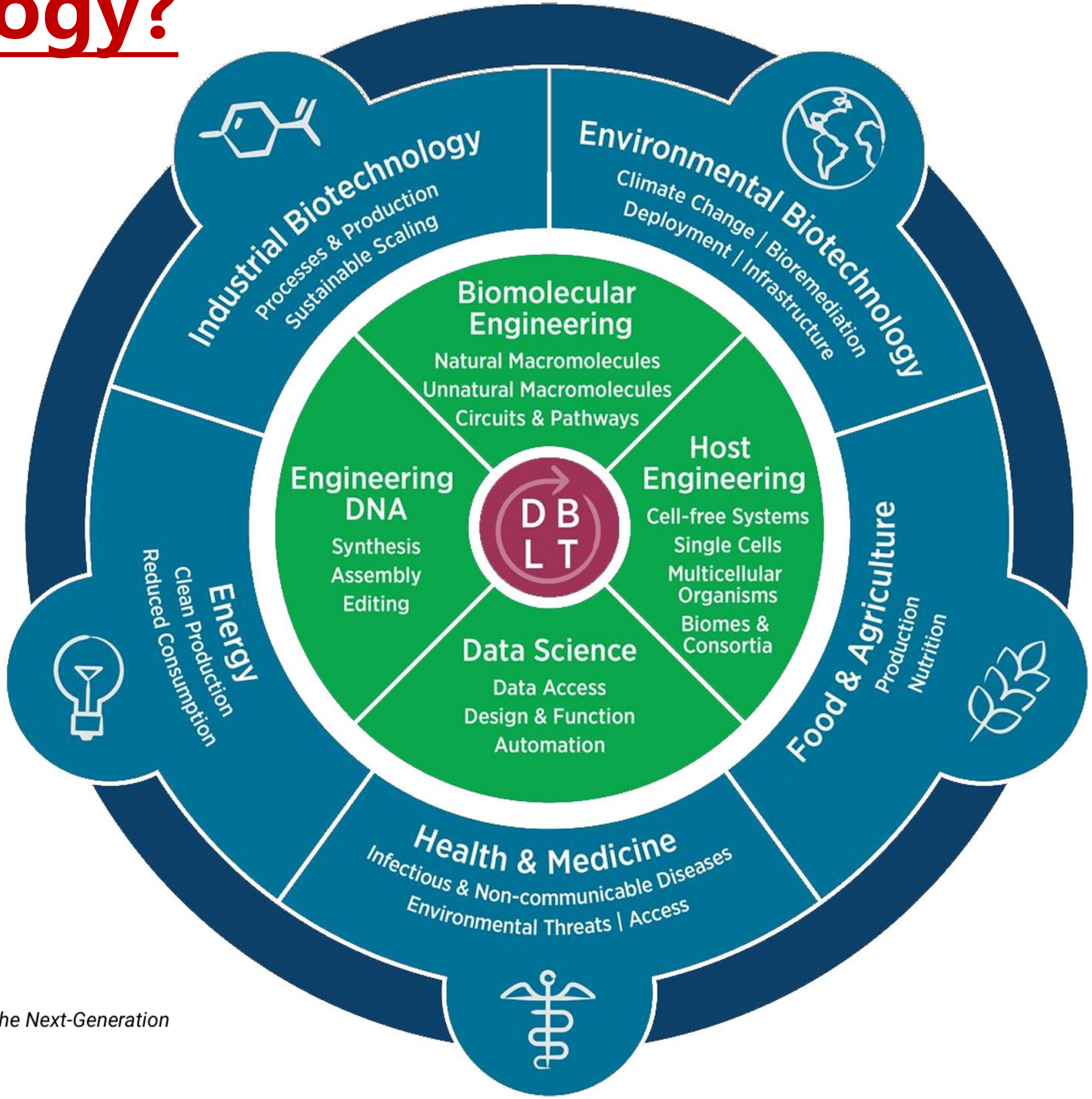
Hence, the overwhelming physical details of natural biology (gene sequences, protein properties, biological systems) must be organized and recast via a set of design rules that hide information and manage complexity, thereby enabling the engineering of many-component integrated biological systems.

It is only when this is accomplished that designs of significant scale will be possible.

上海科技大学

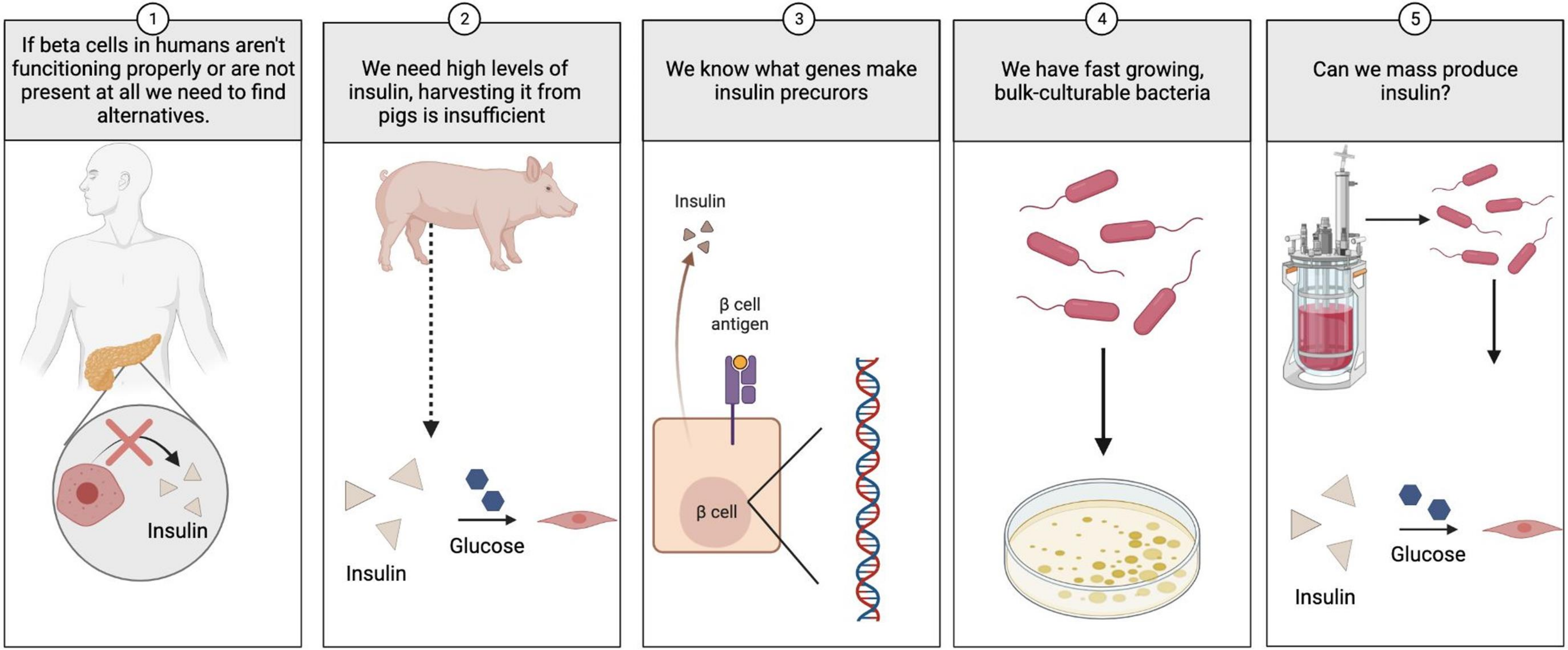
What is Synthetic Biology?

The goal: To "engineer living cells to do something useful; for example, treat a disease, sense a toxic compound in the environment, or produce a valuable drug." (BioBuilder)



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from http://roadmap.ebrc.org. doi: 10.25498/E4159B.

Industrial: Bacterial production of insulin



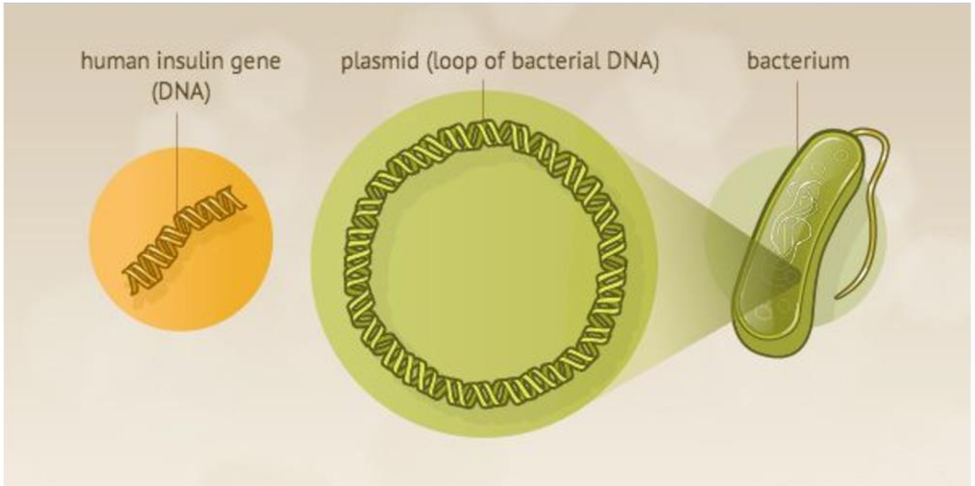
Mona Dabbas & Javin Oza, California Polytechnic State University.

Industrial: Bacterial production of insulin

上海科技大学 ShanghaiTech University

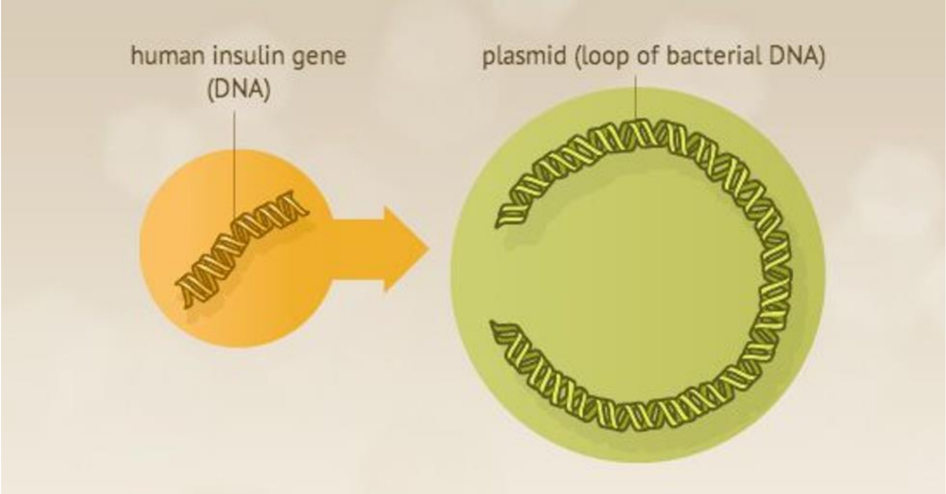
What tools do we need?

- Insulin precursor genes
- Genetic parts for gene expression in *E. coli*
- Way to assemble these genetic parts
- The *E. coli* cells themselves
- Way to get the DNA into the cells
- Way to harvest insulin from bacteria
- Way to measure insulin production



What steps do we need to take?

- Design a genetic circuit that will produce insulin in *E. coli*
- Build the genetic circuit, and put it into bacteria
- Test how much insulin is made
- Learn what did/didn't work, change design accordingly

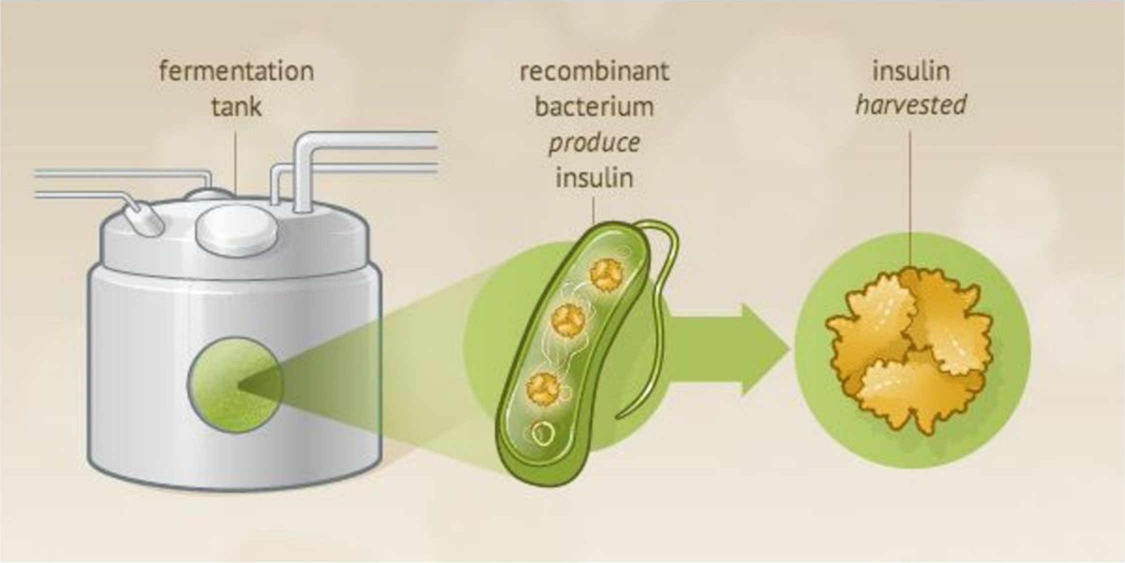


NIH. From DNA to Beer: Harnessing Nature in Medicine & Industry. 2013

Industrial: Bacterial production of insulin

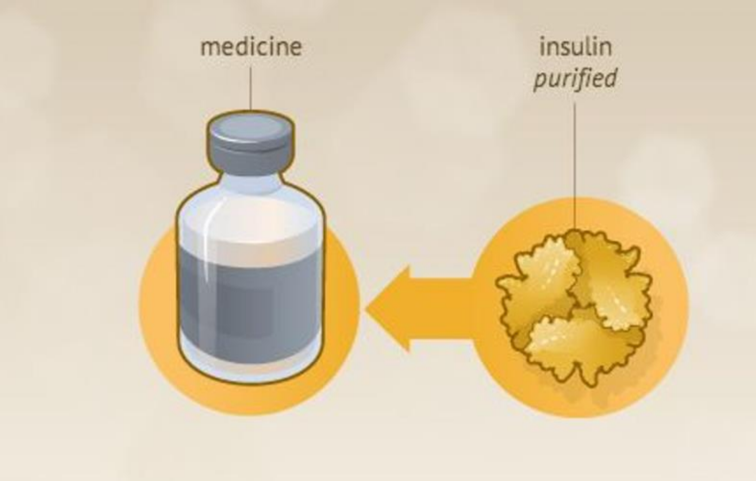
What tools do we need?

- Insulin precursor genes
- Genetic parts for gene expression in *E. coli*
- Way to assemble these genetic parts
- The *E. coli* cells themselves
- Way to get the DNA into the cells
- Way to harvest insulin from bacteria
- Way to measure insulin production



What steps do we need to take?

- Design a genetic circuit that will produce insulin in *E. coli*
- Build the genetic circuit, and put it into bacteria
- Test how much insulin is made
- Learn what did/didn't work, change design accordingly



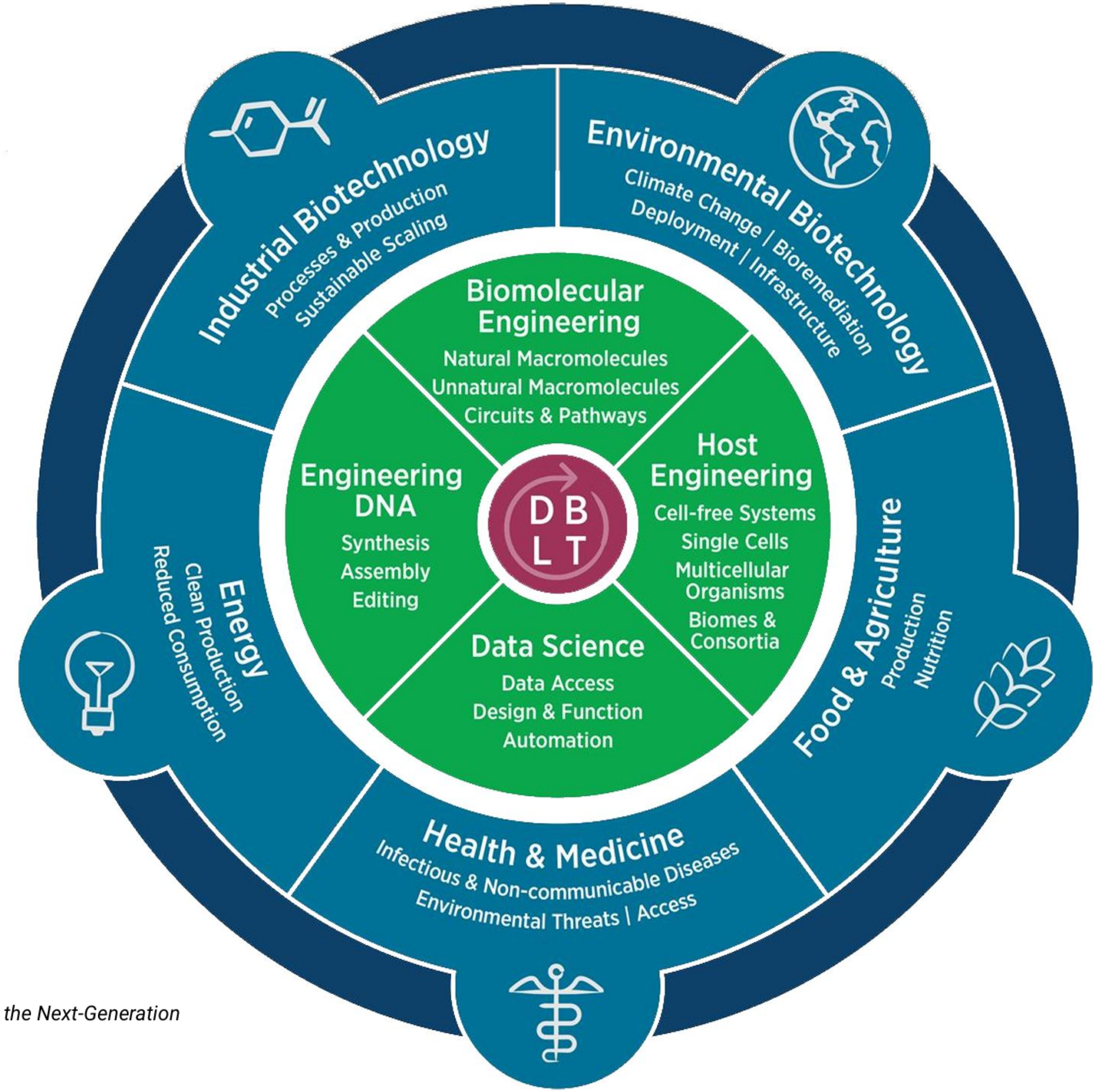
NIH. From DNA to Beer: Harnessing Nature in Medicine & Industry. 2013

SynBio Roadmap

From a Design, Build, Test, Learn centered approach

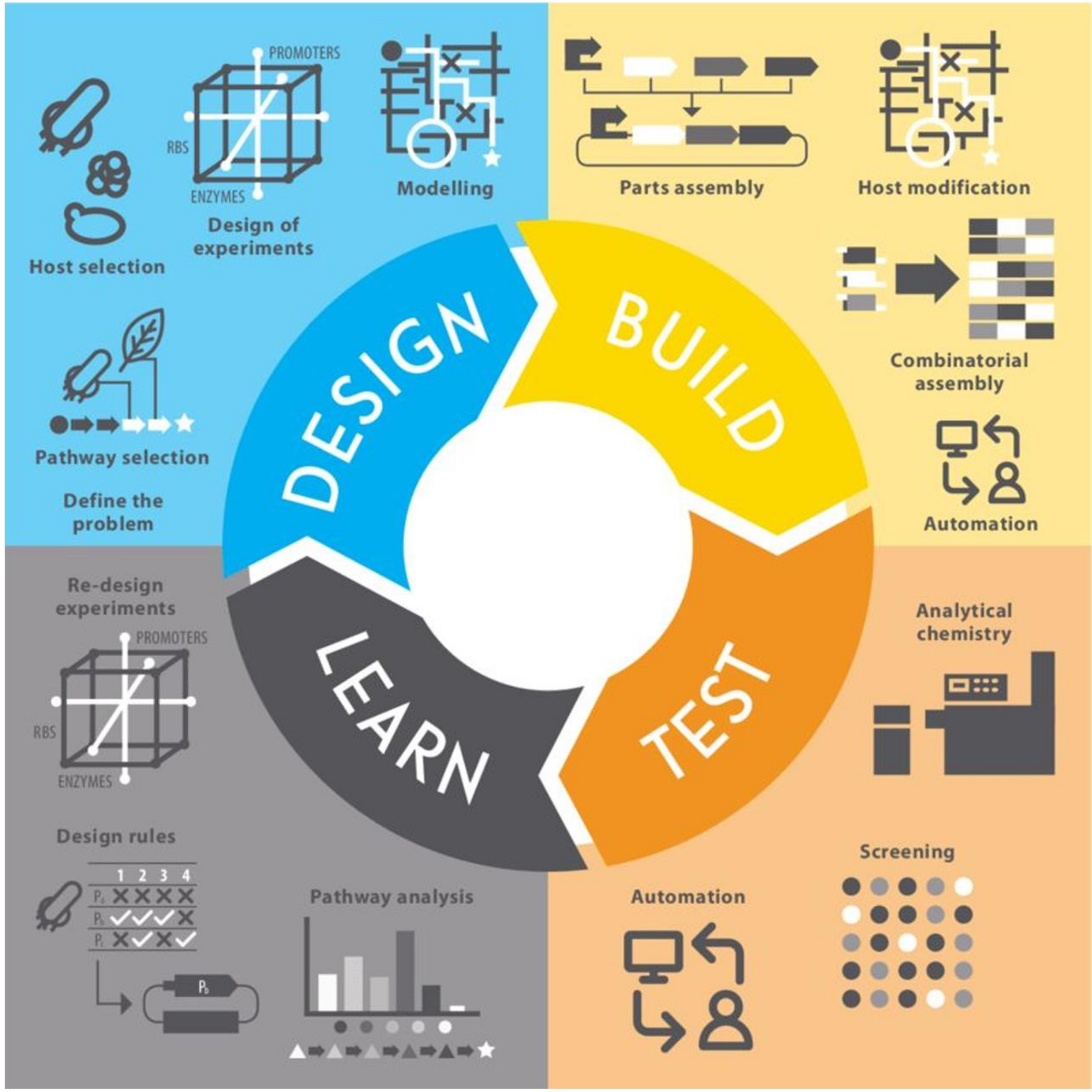
Uses biological, engineering, and computational tools

To positively impact multiple sectors



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from http://roadmap.ebrc.org. doi: 10.25498/E4159B.

The DBTL Cycle



Design a biological circuit or system for a specific function using literature & computational tools.

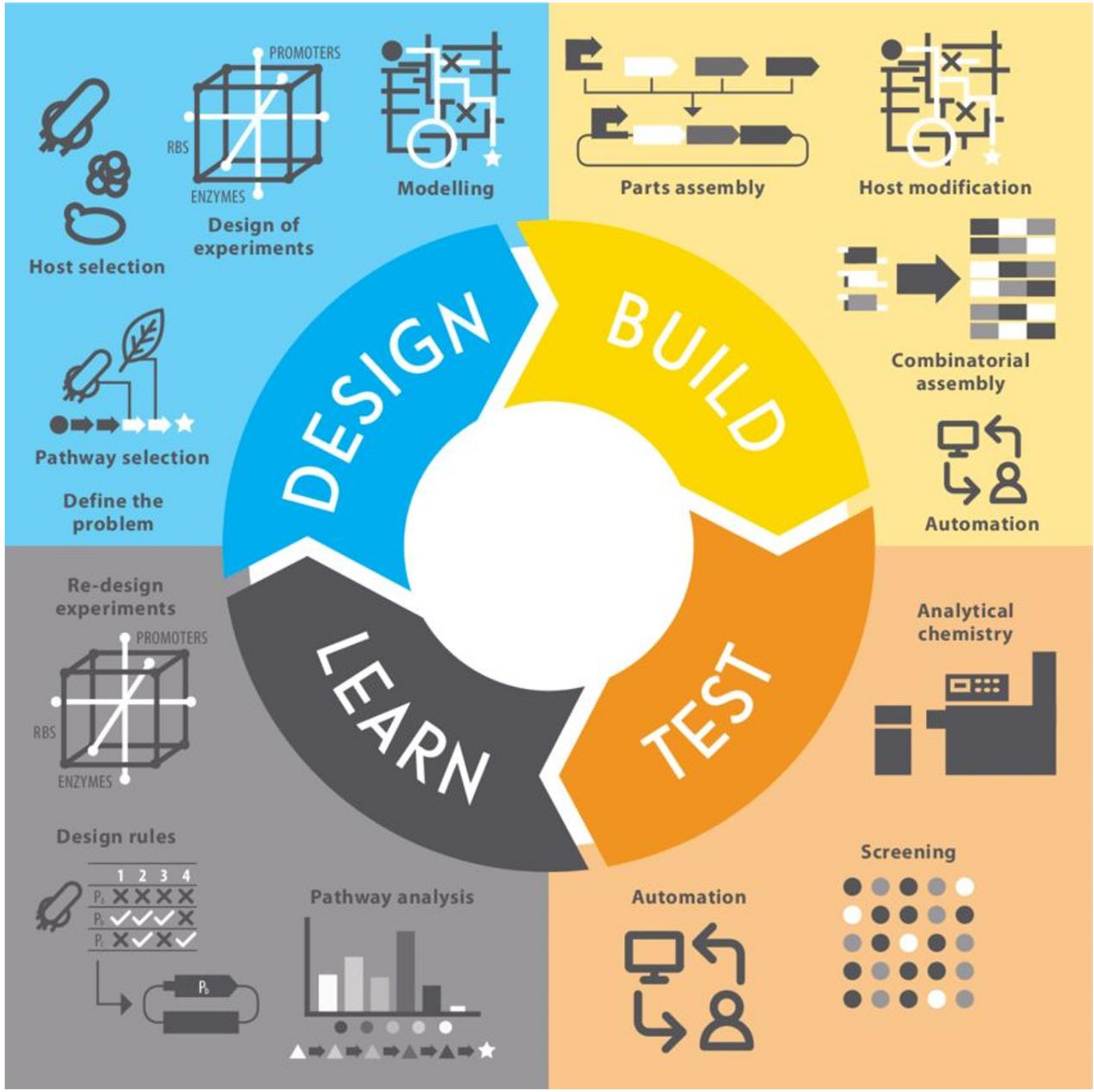
Build your system by assembling, editing, and installing genetic parts.

Test system functionality using robust metrics & controls.

Learn from how the system (doesn't) work to create new design rules, and share your knowledge.

Gray, P. et al. *Synthetic Biology in Australia: An Outlook to 2030* (Australian Council of Learned Academies, 2018).

DBTL: In Practice



Design by finding genes for human insulin precursors, and optimize/model *E. coli* expression.

Build gene sequence including insulin precursor genes though synthesis & assembly.

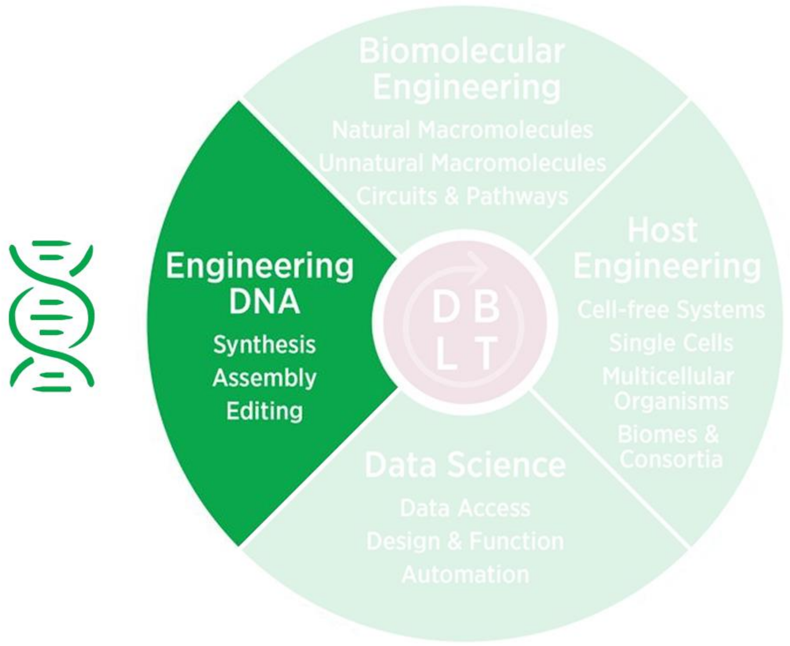
Test your circuit in cells by measuring for insulin precursor proteins chemically via an assay.

Learn how well your system works, and what optimization is necessary.

Gray, P. et al. *Synthetic Biology in Australia: An Outlook to 2030* (Australian Council of Learned Academies, 2018).

Core tools for Engineering DNA

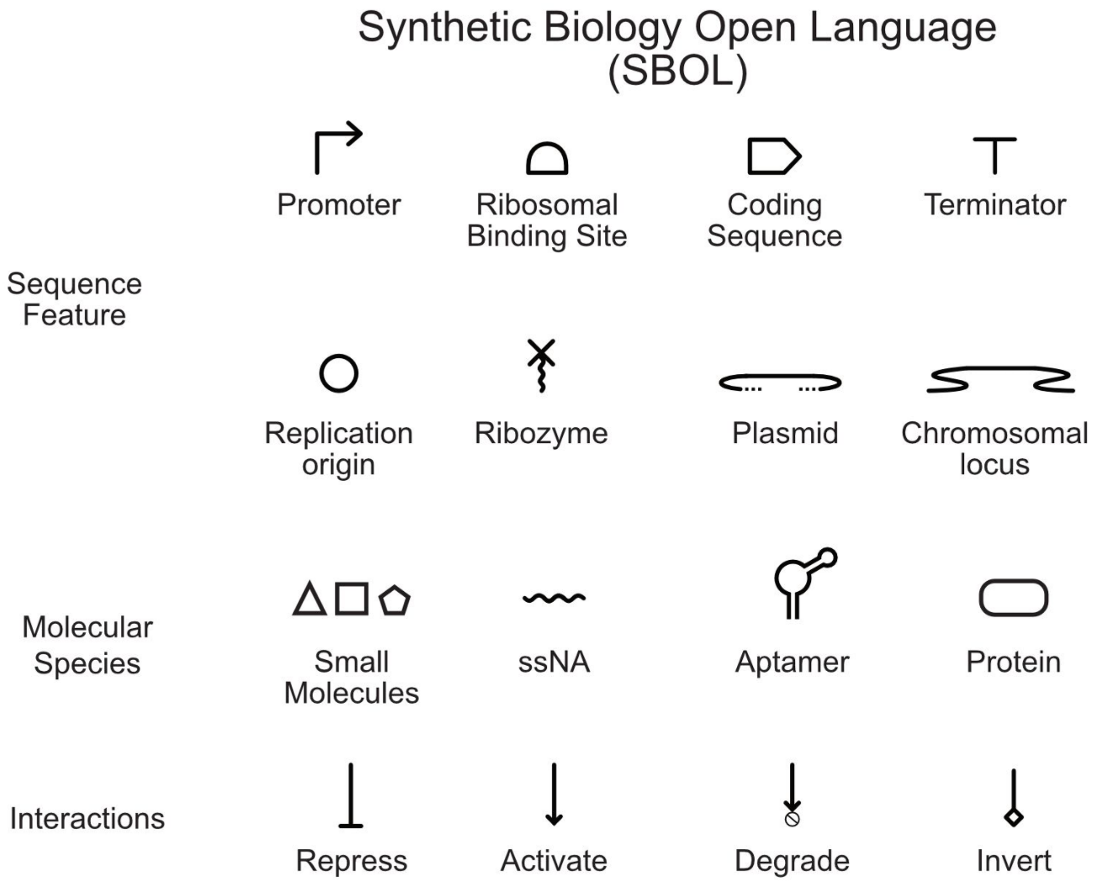
- What are the current tools for engineering DNA?
- Tools enabling a engineering DNA DBTL
  - Synthesis
  - Sequencing
  - Standardization
  - Assembly
  - Editing



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from http://roadmap.ebrc.org. doi: 10.25498/E4159B.

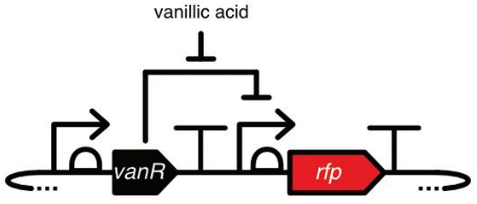
Engineering DNA: Standardized depiction of gene circuits

- Engineering Biology often requires descriptions of the DNA sequences used and how they function
- With the wealth of easy to synthesize DNA parts how do we communicate what pieces of synthetic DNA do?
- The Synthetic Biology Open Language (SBOL) is a framework for describing synthetic DNA sequences and their functional relationships

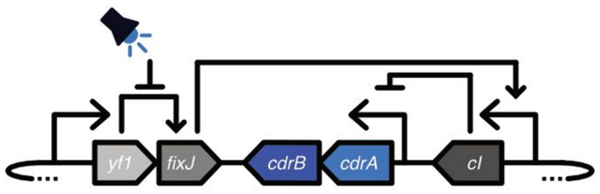


Engineering DNA: Examples of SBOL-based diagrams

- On the left is a plasmid for chemically-inducible expression red fluorescent protein
- On the right is a plasmid for the light-inducible expression of the adhesins CdrAB
- Both of these plasmids express components that control the expression of other components
- The logical interpretations of the diagrams are listed below the image.



if(vanillic acid) :
    express(rfp)

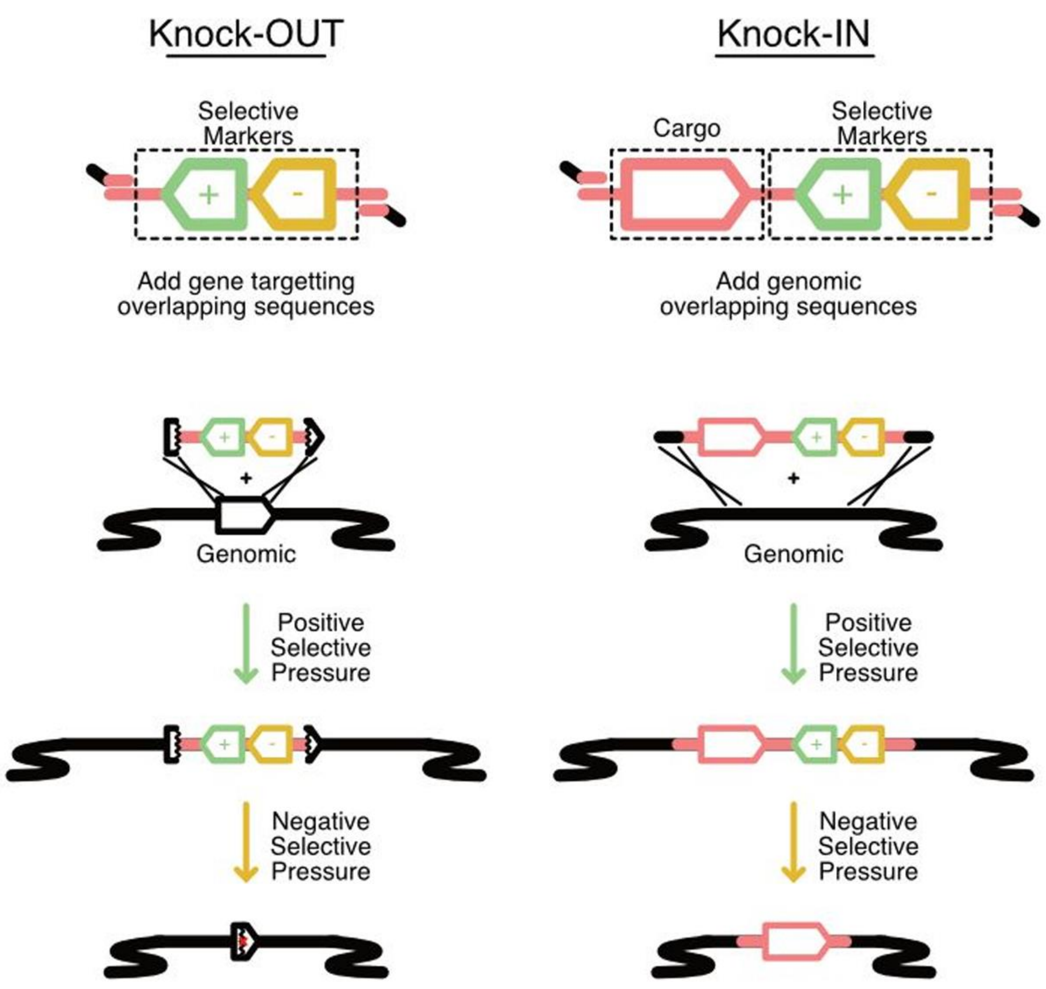


if(blue light):
    inhibit(yf1)
if(yf1):
    activate(fixJ)
if(fixJ-active):
    express(cI)
if(no-cI):
    express(cdrAB)

Engineering DNA: Editing genome by knocking-IN or knocking-OUT

上海科技大学
ShanghaiTech University

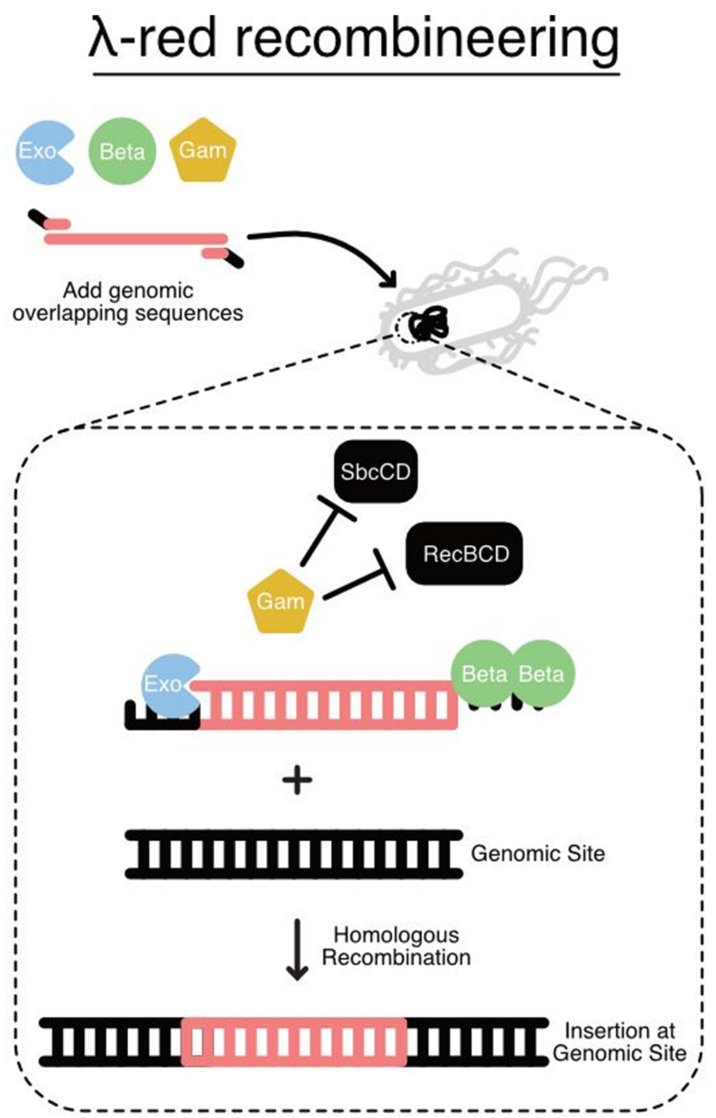
- Genomes can also be engineered
- One strategy for genome editing is to leverage homologous recombination to delete genes (Knock-OUT) or to introduce entirely new sequences (Knock-IN)
- This can be achieved by introducing selective markers to select for cells within populations that have either gained (positive selection) or lost the engineered DNA (negative selection)



立志成才报国裕民

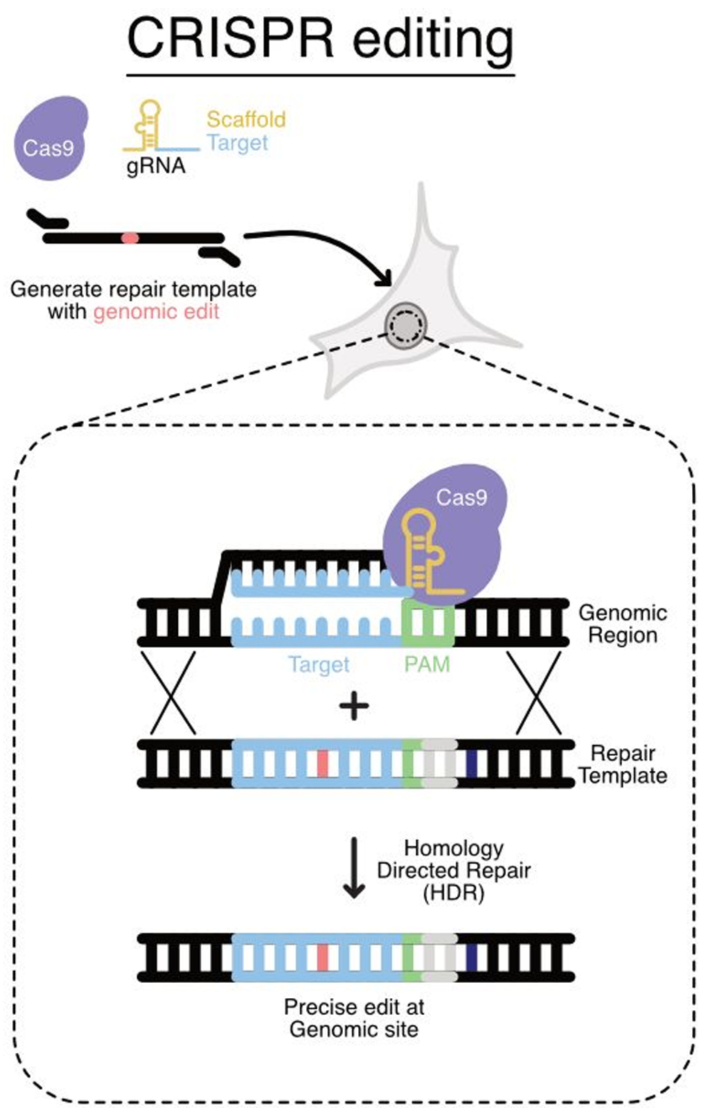
Engineering DNA: Using Recombineering to edit genomes

- Viral mechanisms for DNA integration have been co-opted to increase genomic engineering efficiency
- One approach, called 'recombineering', leverages the *E. coli* lambda prophage (Yu D, et al., *PNAS* (2000), 97(11):5978-5983)
- Prophage proteins Exo, Beta, and Gam improve integration of DNA harboring genomic overlapping sequences by chewing back dsDNA to generate ssDNA, protecting ssDNA from degradation, and blocking host exonucleases, respectively



Engineering DNA: Using CRISPR-Cas9 to make precise genome edits

- Adaptive-immunity mechanisms from bacteria have been co-opted to enable precise genomic edits
- Clustered Regularly Interspaced Short Palindromic (CRISPR) systems from diverse bacteria have enabled a revolution in genomic editing
- Cas9 is one of the most widely used CRISPR systems used for genome engineering (Jinek M., et al., *Science* (2012) 337(6096):816-821)
- Cas9 is an RNA-guided endonuclease that makes double-stranded breaks at specific sites.
- Guide RNAs (gRNA) can be engineered target Cas9 to specific DNA sequences that precede PAM sites (NGG).
- One way to make genomic edits leverages homology directed repair to integrate an engineered repair template that eliminates the PAM site and subsequent retargeting.



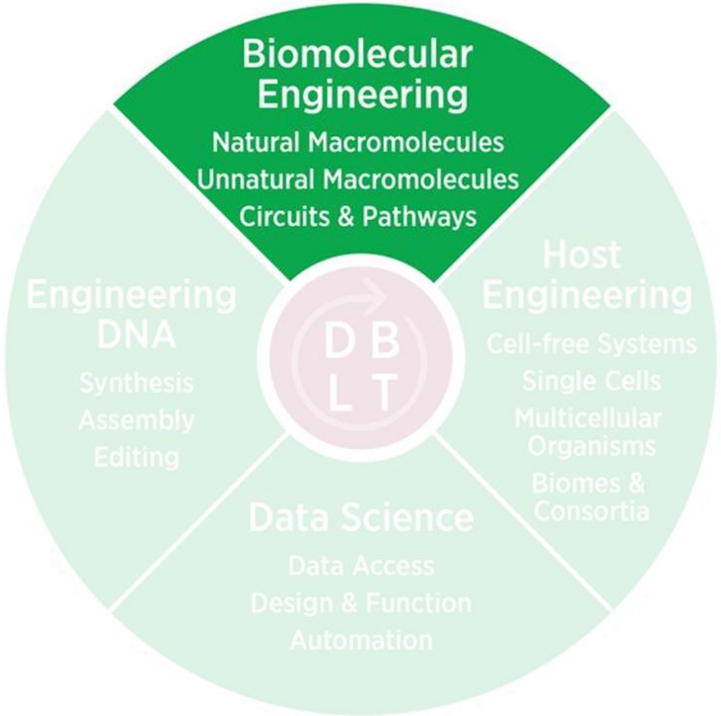
CRISPR gene editing: mechanisms and applications

https://www.bilibili.com/video/BV15x411Z7kf/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4

Core tools for Biomolecular Engineering

上海科技大学 ShanghaiTech University

- What are our current tools for Biomolecular Engineering?
- Tools enable a Biomolecular Engineering DBTL
  - Comprehensive Mutagenesis libraries
  - Functional Screens
  - Functional Selections
  - Directed Evolution

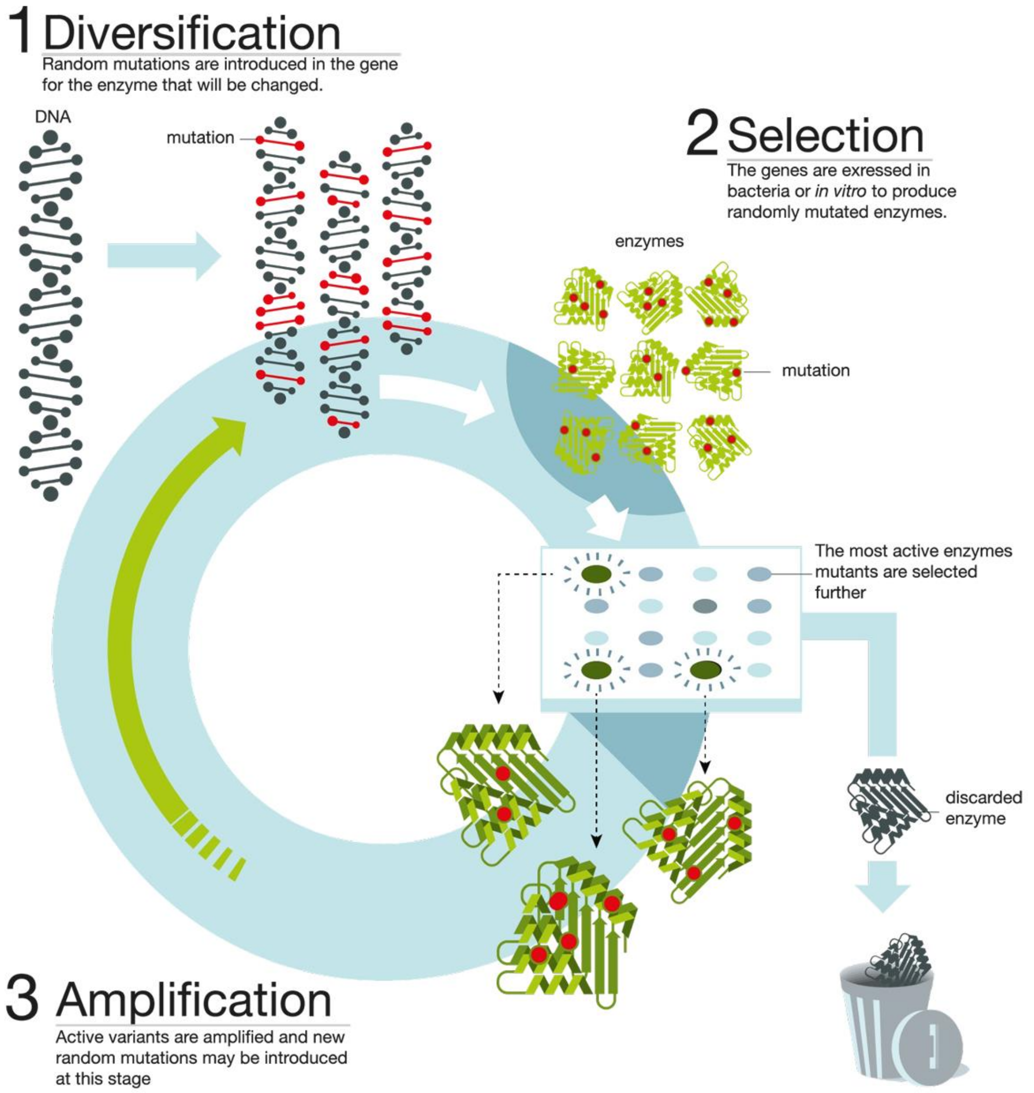


Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from http://roadmap.ebrc.org. doi: 10.25498/E4159B.

立志成才 报国裕民

Biomolecular Engineering: Natural macromolecules

- Natural macromolecules including proteins and RNAs can be engineered to have new properties (*e.g.*, thermostability) or perform new functions (*e.g.*, binding, catalysis)
- Directed evolution can be used to tailor the function of proteins through rounds of sequence diversification, selection for function, and amplification of

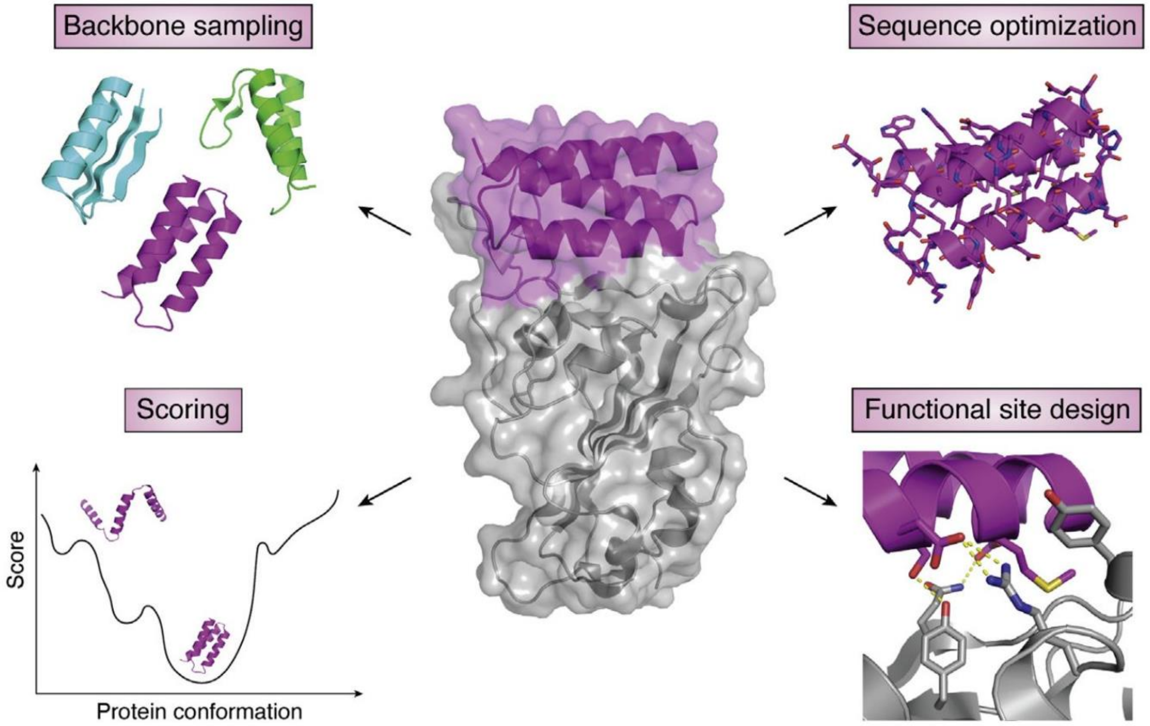


©Johan Jarnestad/The Royal Swedish Academy of Sciences

[The Nobel Prize.](#)

Biomolecular Engineering: Unnatural macromolecules

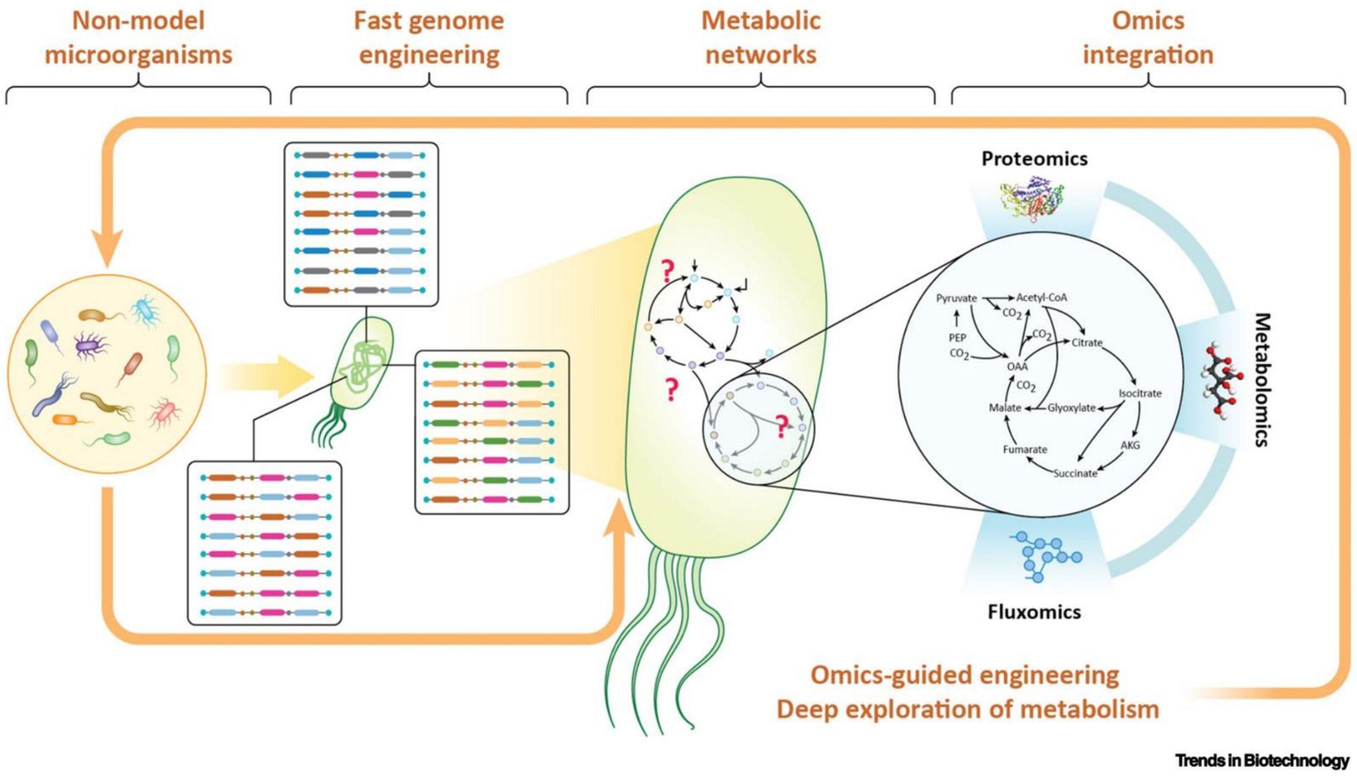
- Entirely new protein sequences not seen in nature can be generated through *de novo* protein design
- *De novo* design enables exploration of protein sequence space beyond what nature has itself explored



Pan X. and Kortemme T. *JBC Reviews* (2021) 296 100558

Biomolecular Engineering: Circuits and Pathways

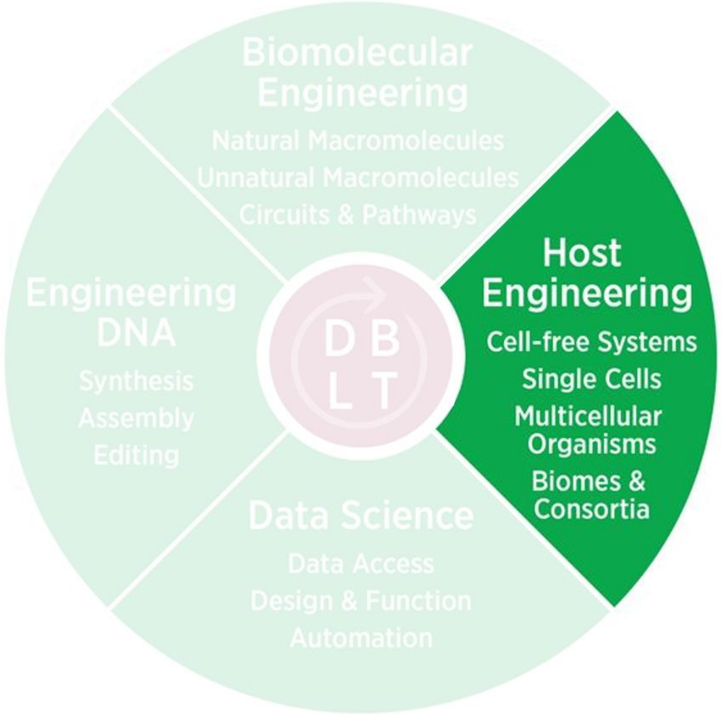
- In addition to individual biomolecules large scale networks of biomolecules
- Novel metabolic pathways can be constructed using retrosynthesis approaches
- Combinatorial libraries can be used to optimize expression of pathways proteins
- Adaptive laboratory evolution can be used to optimize the host genome for improved yield of a target molecule



Gurdo N., Volke D.C., Nikel P.I. (2022) *Trends in Biotechnology*

Host engineering

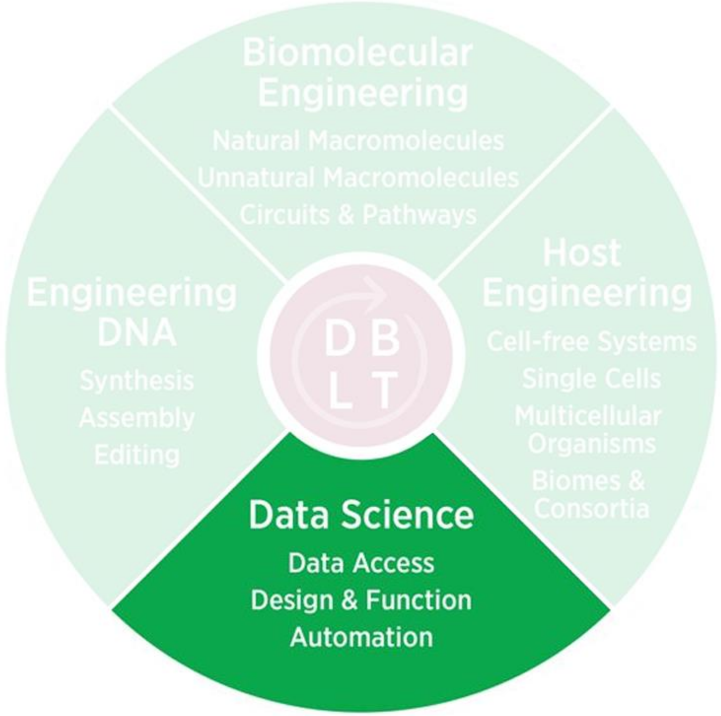
- What types of hosts are there?
- Model & non-model systems
- Thinking across scales



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from http://roadmap.ebrc.org. doi: 10.25498/E4159B.

Core tools for Data Science

- What data is out there?
- Modeling & Machine Learning
- Automation



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from http://roadmap.ebrc.org. doi: 10.25498/E4159B.

Data science: Omics data

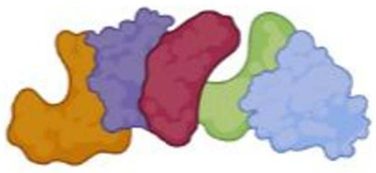
Genome
Who is there?
What can they make?

- Species identification
- Mutation rate in a population

Transcriptome
What are they saying to making?

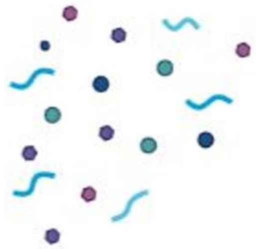
- Gene regulation
- Population heterogeneity

Proteome
What are they making?



- RNA regulation
- Protein stability & degradation

Metabolome
What are they doing?



- Metabolic pathway efficiency
- Cellular inputs & outputs

Then integrate!

Data science: Databases

- National Center for Biotechnology Information (NCBI)
  - GenBank - NIH genetic seq. collection
  - Nucleotide BLAST - search gene seqs.
- UniProt
  - Protein seq. and functional information
- RCSB Protein Database (PDB)
  - Characterized protein structure and models
- Other, more specific DBs:
  - FPBase - fluorescent protein characterization and lineages

Data science: Modeling genetic parts

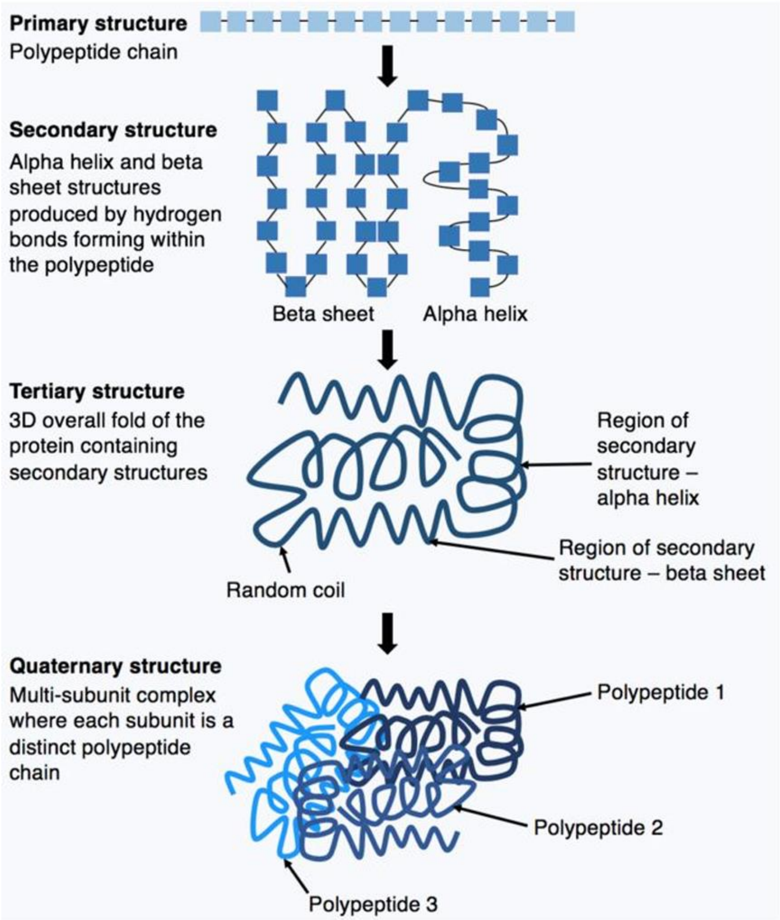
- Can use first principles thermodynamics to predict function of parts in some cases
- Can integrate larger datasets with machine learning models to make more sophisticated predictions

De Novo DNA

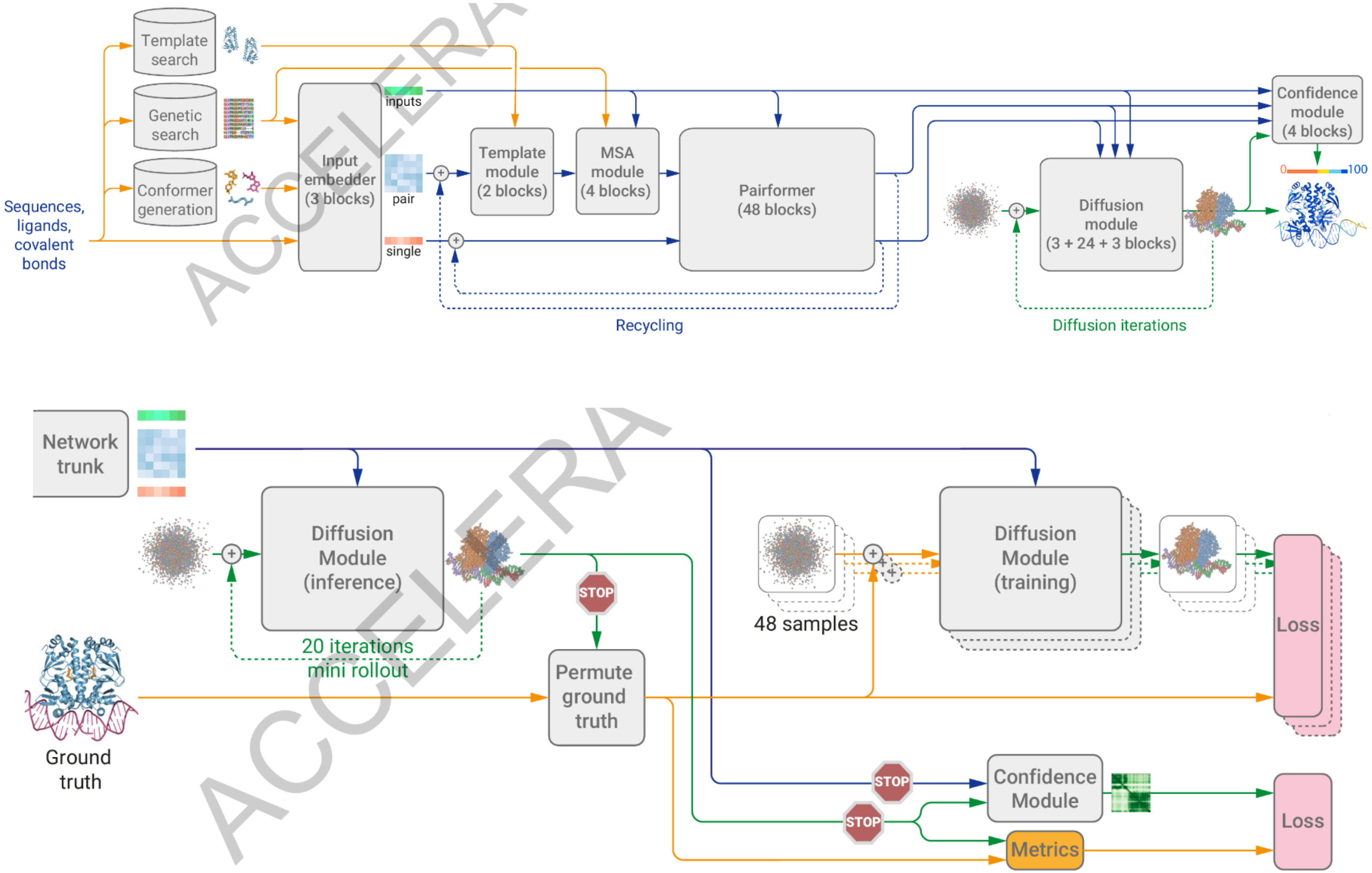
Predict RBS output based on both RBS and protein sequence

Benchling

Data science: Modeling proteins (AlphaFold)



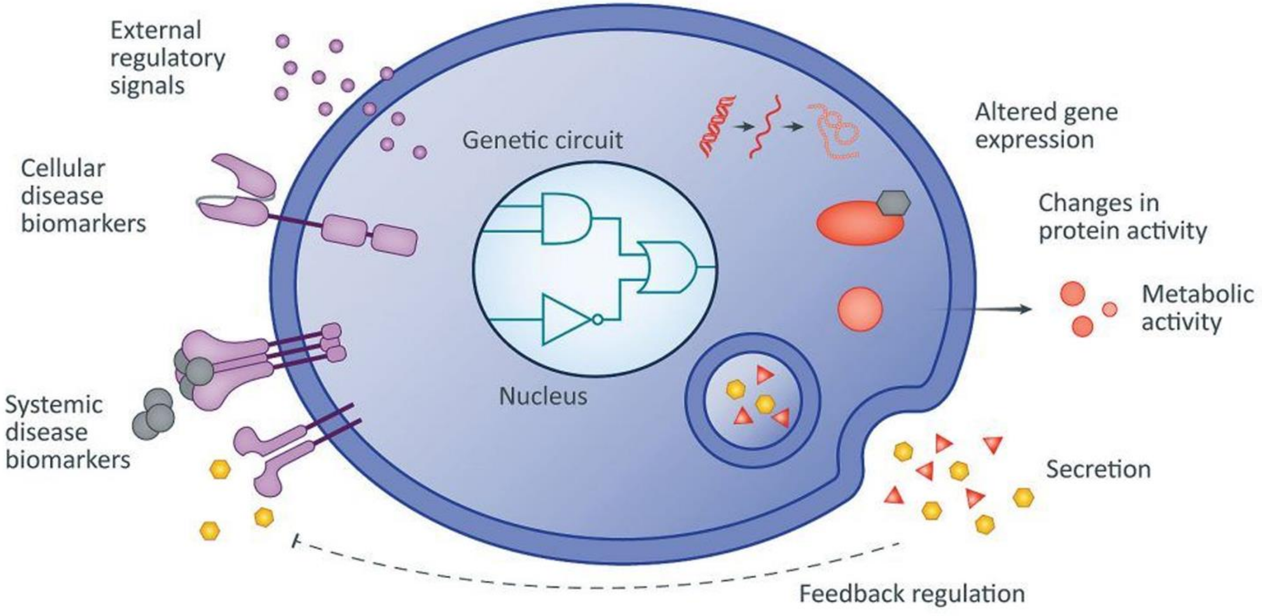
[Kep17, "Amino acid chains."]



AlphaFold3, Google DeepMind

Data science: Modeling whole systems

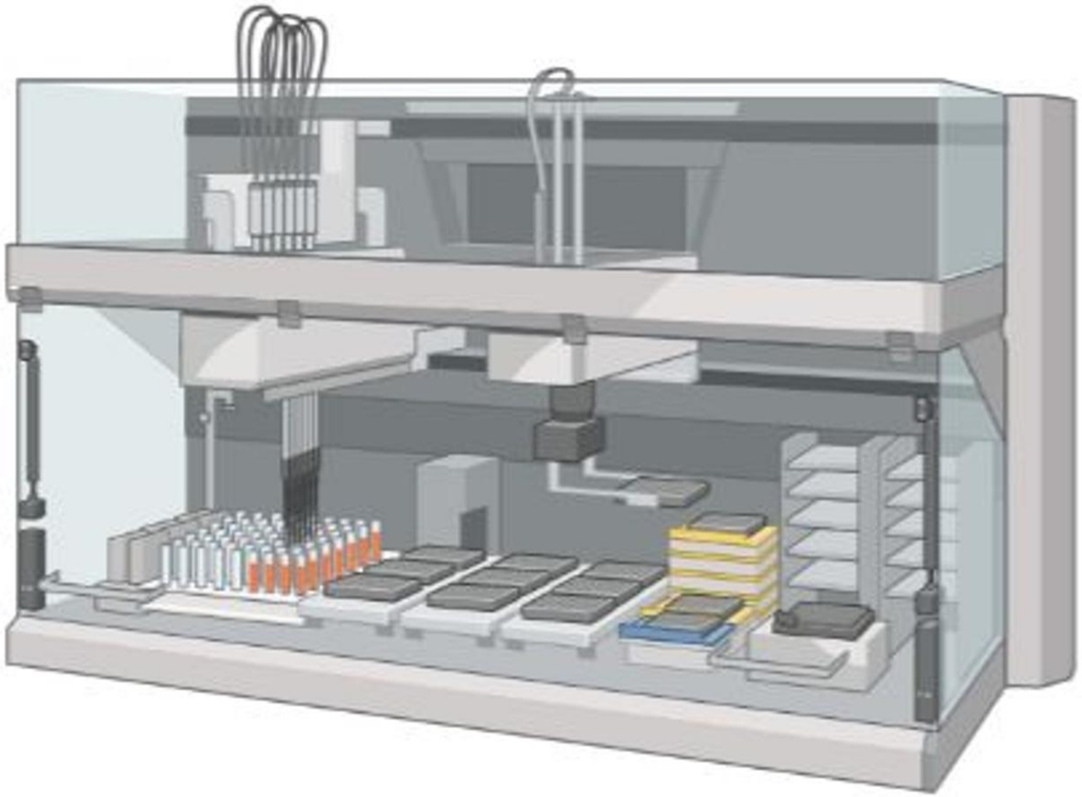
- Quantify what is there (-omics)
- Determine inputs and outputs
- Apply equations to determine system functions - metabolic efficiency, protein activity, etc.
- NIH: ‘Systems biology is an approach to understand the larger picture—be it at the level of the organism, tissue, or cell—by putting its pieces together. ‘



[Lu, T. (2020). Senti Biosciences.](#)

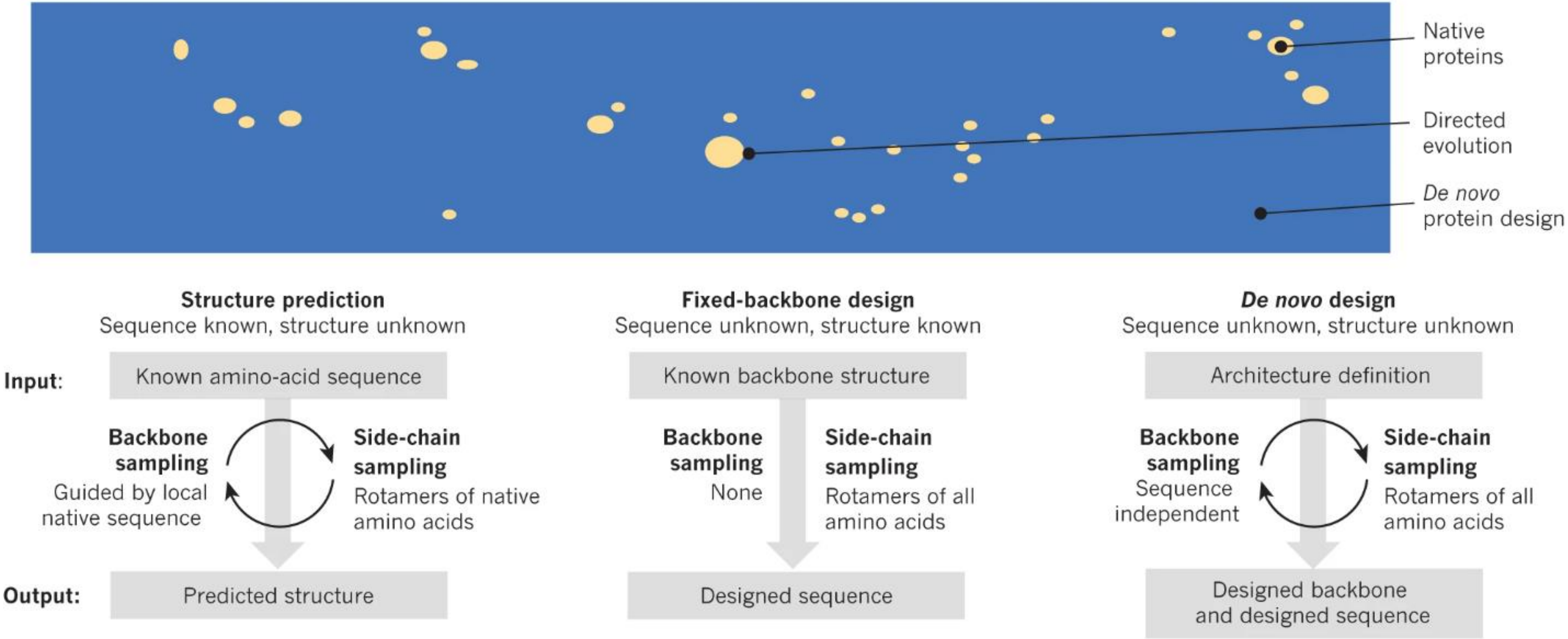
Data science: Automation of laboratory processes

- Liquid handlers to automate Building & Testing genetic constructs
- Increase throughput to test multiple constructs simultaneously
- Integrated analysis for -omics



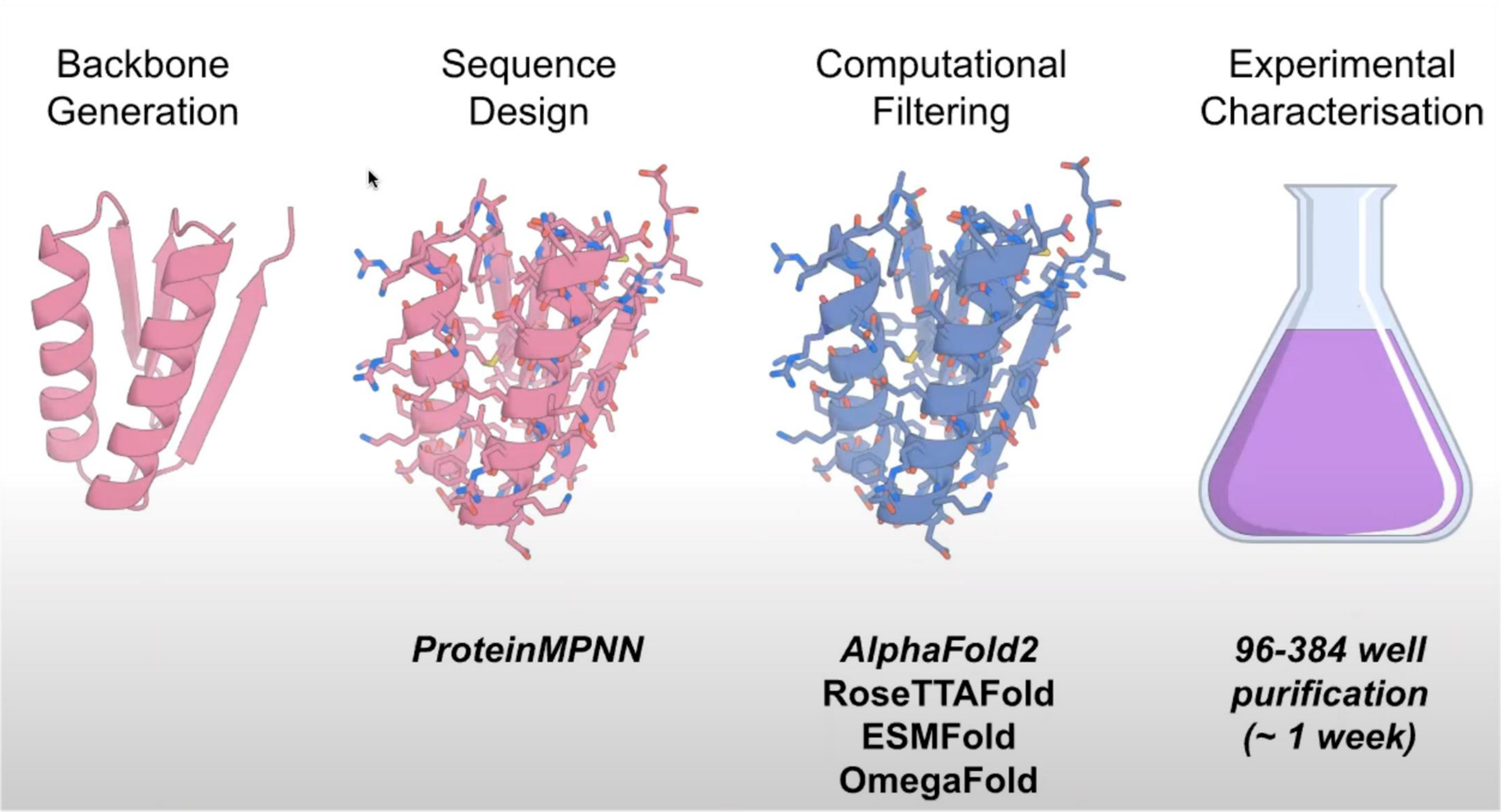
AI for Protein Design

Protein Design



P.-S. Huang, S. E. Boyken, D. Baker. "The coming of age of *de novo* protein design", Nature 2016.

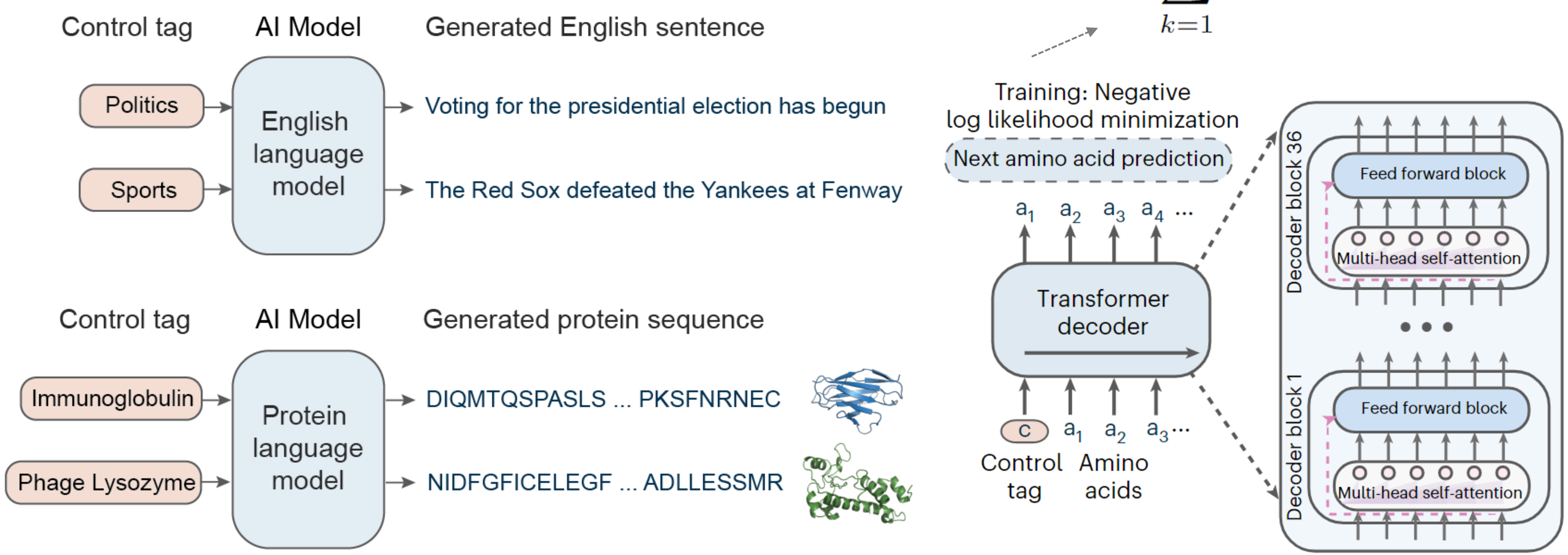
General protein design process



https://www.youtube.com/watch?v=wIHwHDt2NoI

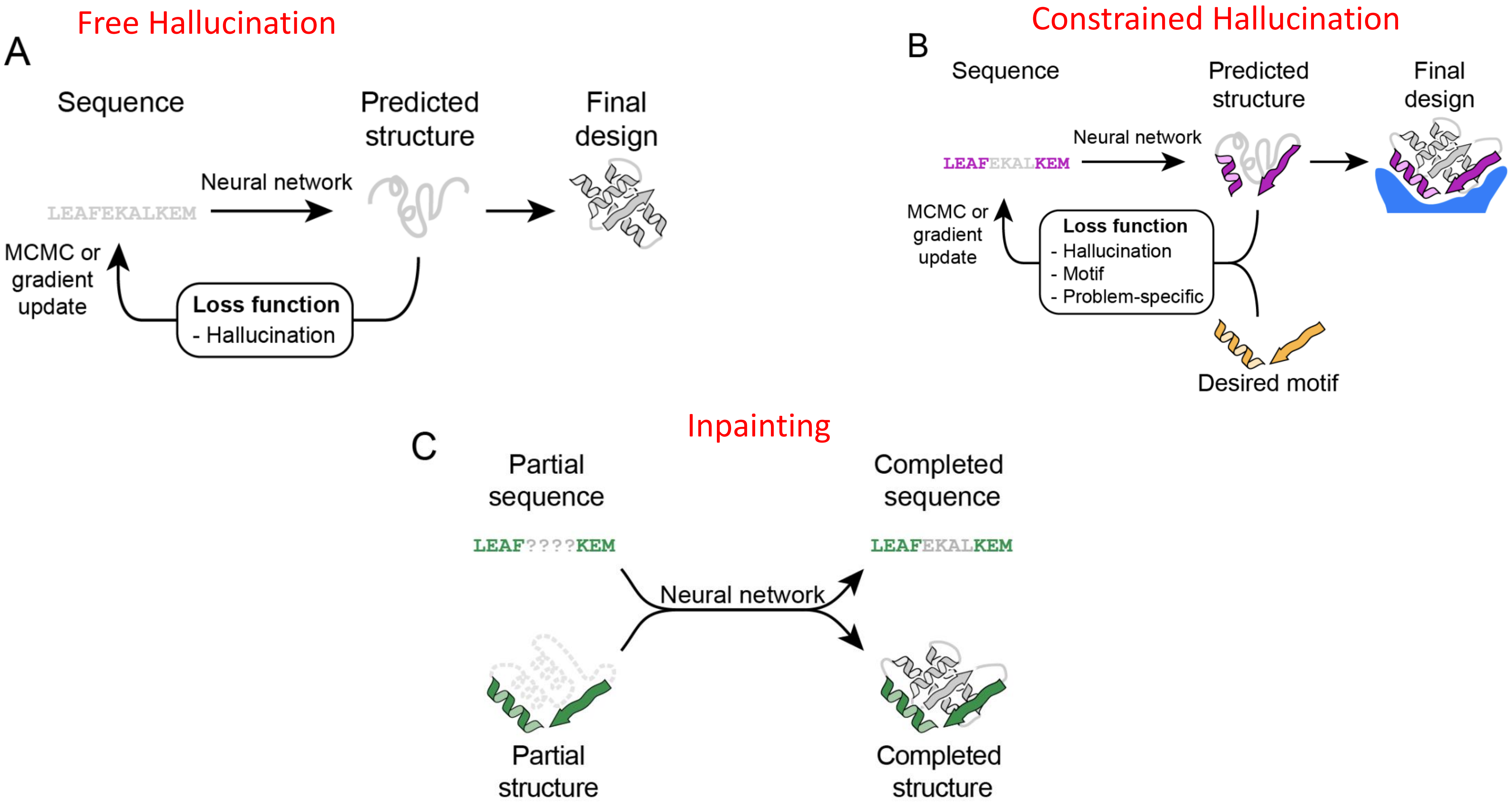
ProGen

$$\mathcal{L}(D) = -\sum_{k=1}^{|D|} \log p_\theta(x_i^k | x_{<i}^k)$$



Ali Madani et al, Nat Biotechnol 2023

Methods for protein function design



J. Wang, S. Lisanza, D. Juergens, D. Tischer, et al. bioRxiv 2021.

Example of training and fine-tuning data

Sample sequences with associated control tags provided during the training and fine-tuning

Template: $<c_1>\ <c_2>\ \ldots\ <c_N>\ a_1\ a_2\ a_3\ a_4\ a_5\ \ldots$

Training Sample

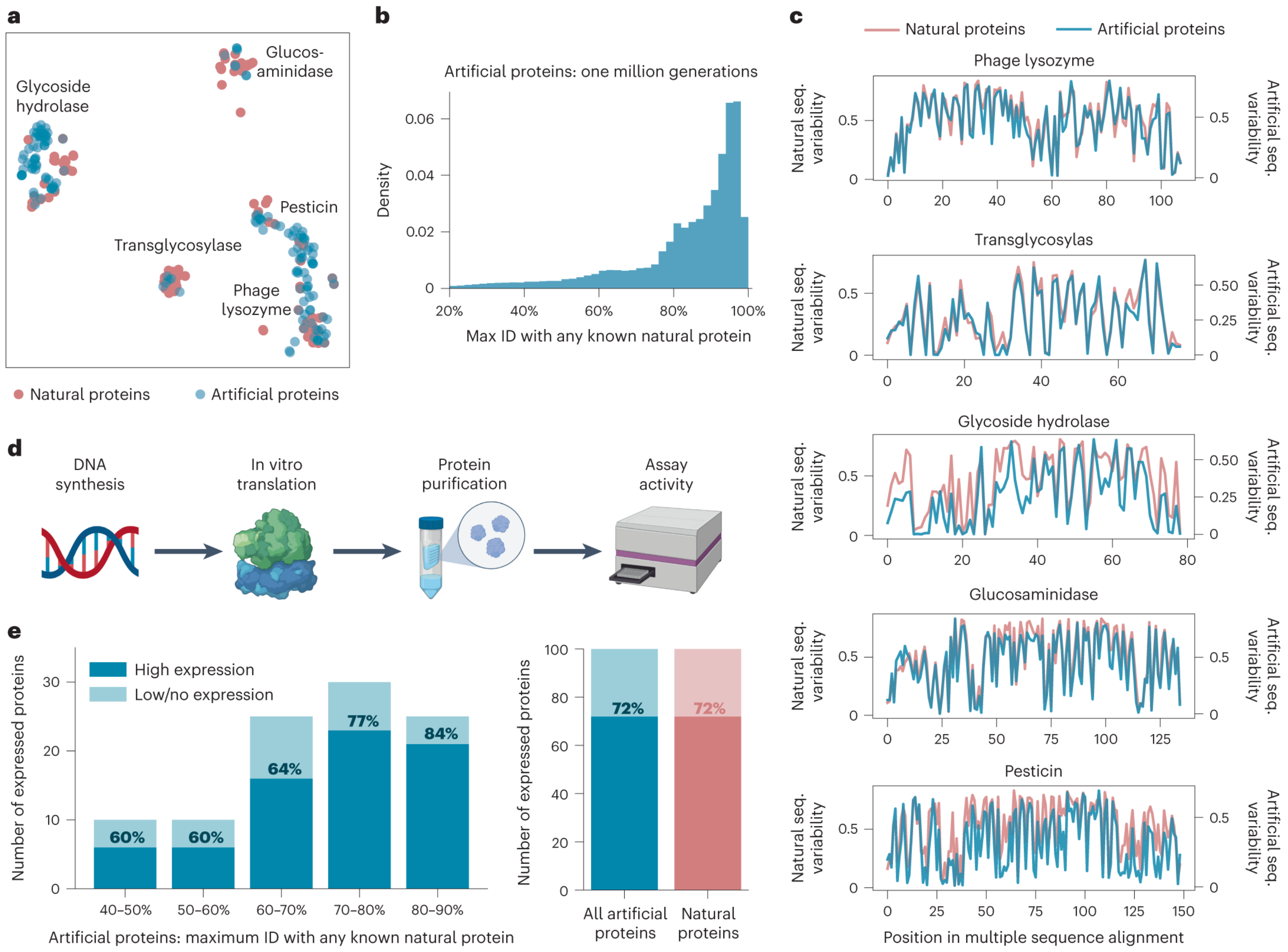
```
<Metazoa><Chordata><Mammalia><Rodentia><Muridae><Rattus><Rattus><Norvegicus>
<NAD>
<Translocase>
<Iron><Iron-sulfur><2Fe-2S>
<Mitochondrion><Mitochondrion inner membrane>
<Transport><Electron transport><Respiratory chain>
MFSLALRARASGLTAQWGRHARNLHKTAVQNGAGGALFVHRDTPENNPDTPFDFTPENYERIEAIVRNYPEGHRAA
AVLPVLDLAQRQNGWLPISAMNKVAEVLQVPPMRVYEVATFYTMYNRKPVGKYHIQVCTTTPCMLRDSDSILETLQ
RKLGIKVGETTPDKLFTLIEVECLGACVNAPMVQINDDYYEDLTPKDIEEIIDELRAGKVPKPGPRSGRFCCEPAG
GLTSLTEPPKGPGFGVQAGL
```

Fine-tuning Sample

```
<Phage Lysozyme>
MNIFEMLRIDEGLRLKIYKDTEGYYTIGIGHLLTKSPSLNAAKSELDKAIGRNCNGVITKDEAEKLFNQDVDAAV
RGILRNAKLKPVYDSLDAVRRCALINMVFQMGETGVAGFTNSLRMLQQKRWDEAAVNLAKSRWYNQTPNRAKRVI
TTFRTGTWDAYKNL
```

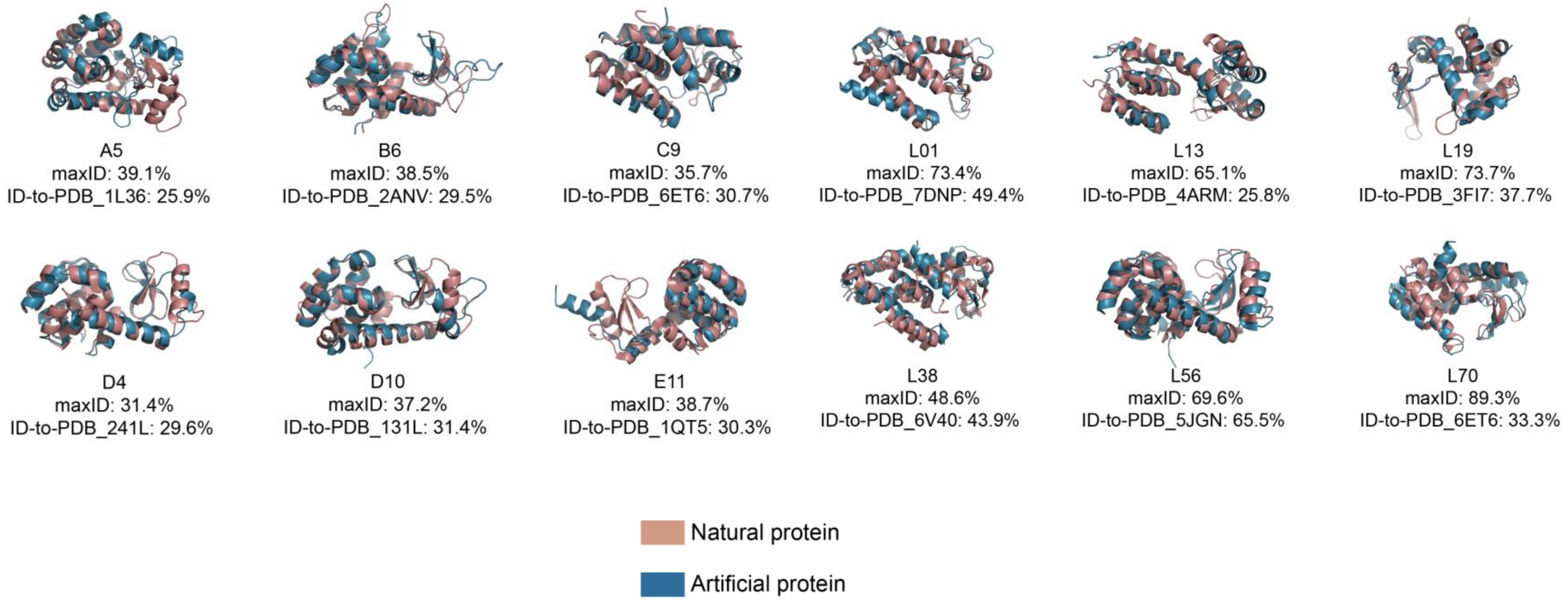
Ali Madani et al, Nat Biotechnol 2023

Generated proteins are diverse and express well



Ali Madani et al, Nat Biotechnol 2023

Artificial proteins are similar to natural proteins in structures



Ali Madani et al, Nat Biotechnol 2023

上海科技大学
ShanghaiTech University

ProGen2: Exploring the Boundaries of Protein Language Models

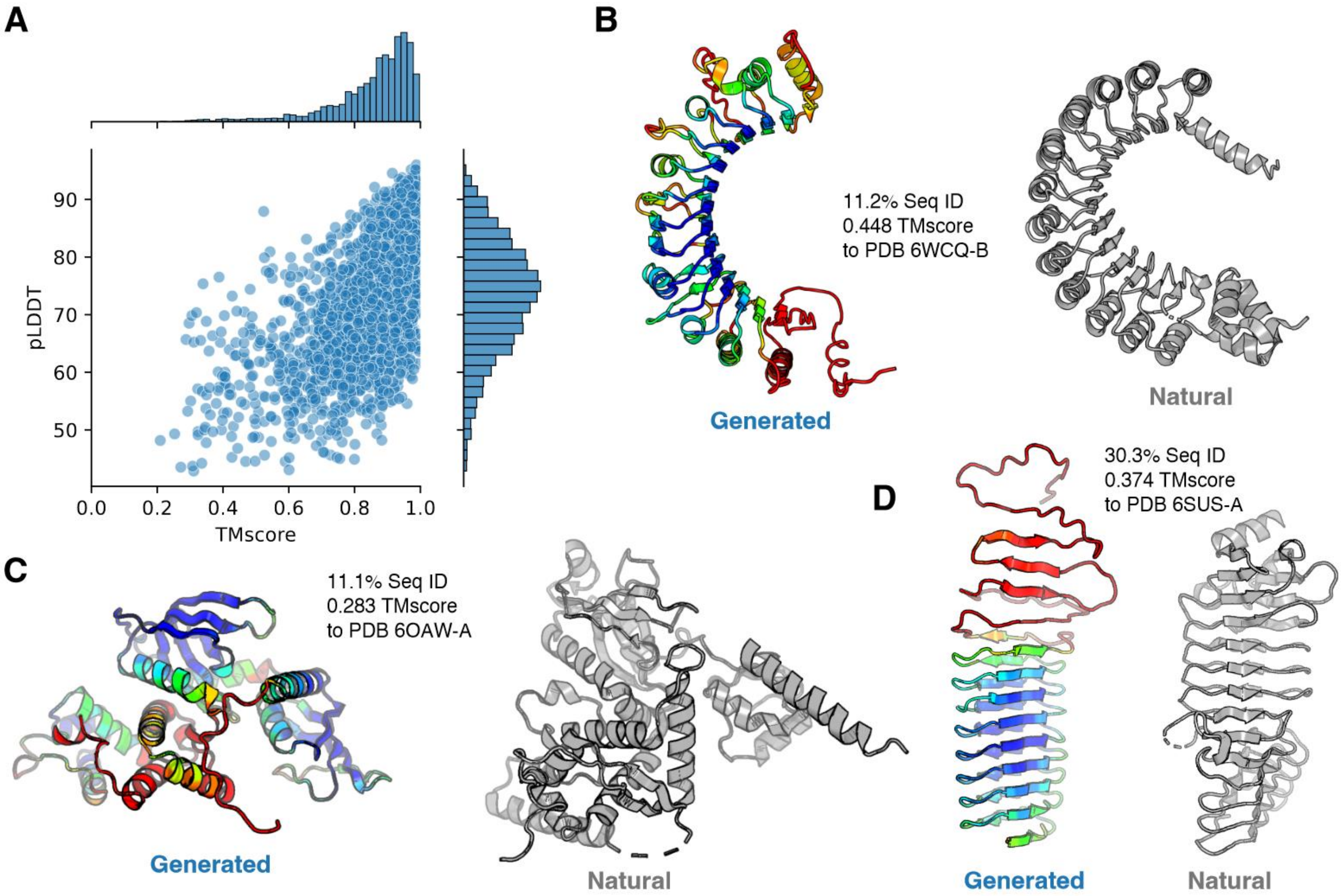
Erik Nijkamp* Salesforce Research
Jeffrey Ruffolo* Johns Hopkins University
Eli N. Weinstein Columbia University
Nikhil Naik Salesforce Research

Ali Madani Salesforce Research

Submitted to arXiv, 27 Jun 2022, https://arxiv.org/abs/2206.13517

立志成才报国裕民

Generation from pre-trained models



Weaknesses of ProGen

- **ProGen1:**
  - The effect of model for a specific protein family is dependent on the quantity and quality of the specific fine-tuning dataset
  - It is time-consuming to add control tags to sequences of the pretrained dataset

- **ProGen2:**
  - ProGen2 cannot generate sequences containing specific structural and functional domains without fine-tuning
  - Compared with current top Large Language Models (LLMs), the model size and dataset size of ProGen2 are still not large enough

RFdiffusion

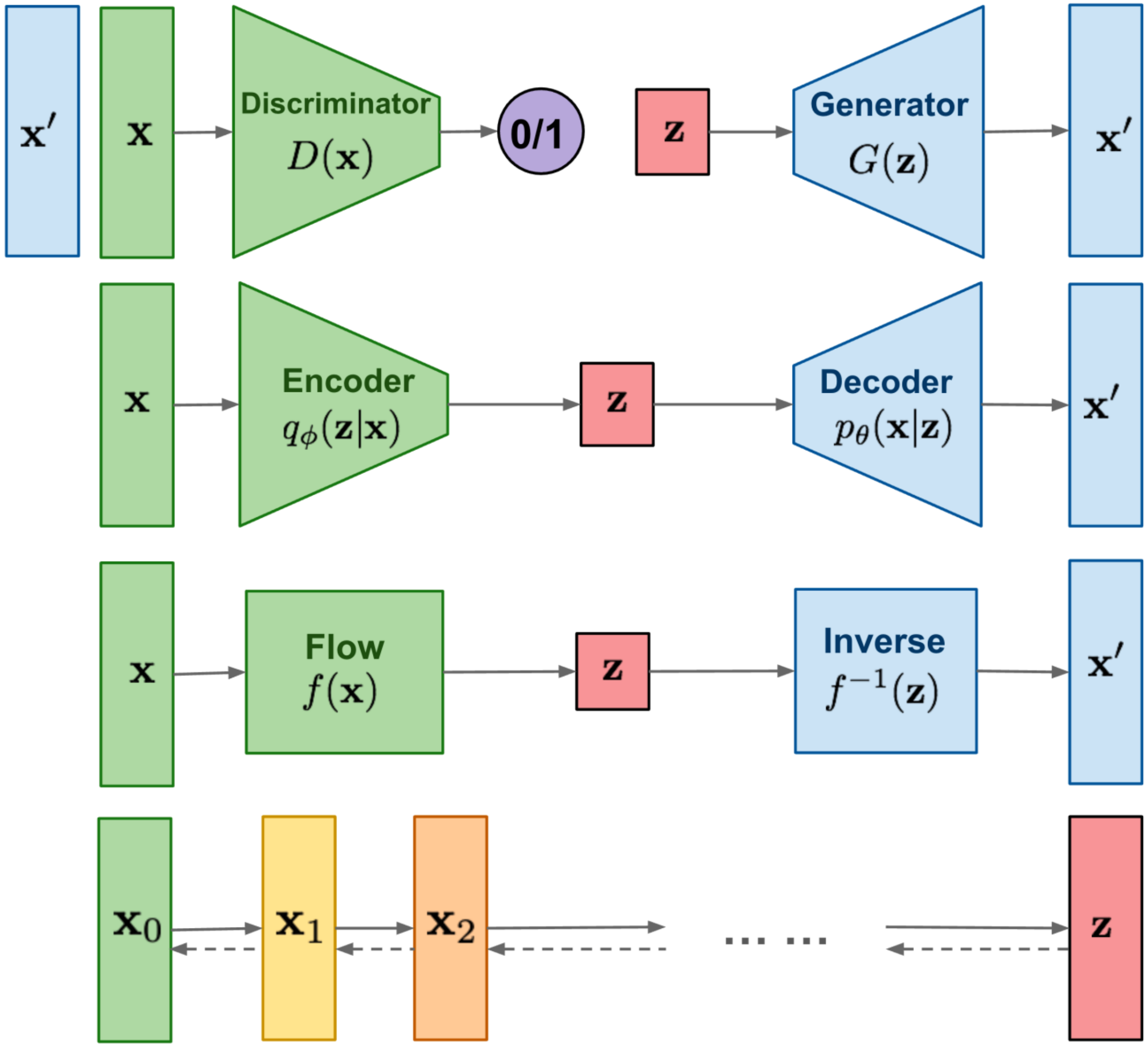
Generative models

GAN: Adversarial training

VAE: maximize variational lower bound

Flow-based models: Invertible transform of distributions

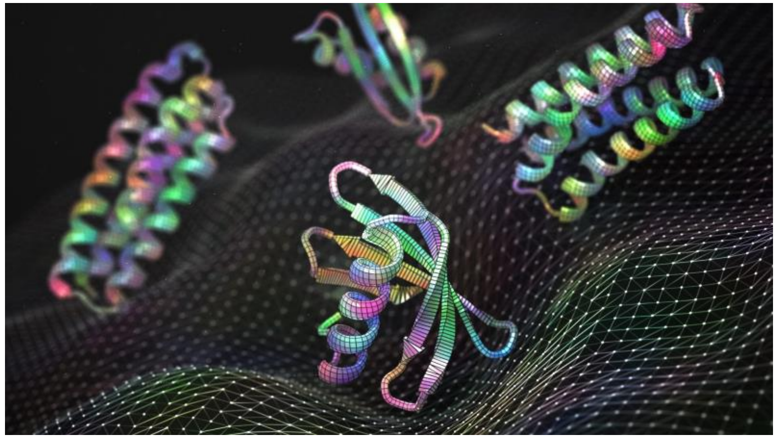
Diffusion models: Gradually add Gaussian noise and then reverse



https://lilianweng.github.io/posts/2021-07-11-diffusion-models

Diffusion in computational biology

后AlphaFold2时代，研究者们努力发掘更大的AF2落地场景，即**从结构预测走向结构设计/生成**

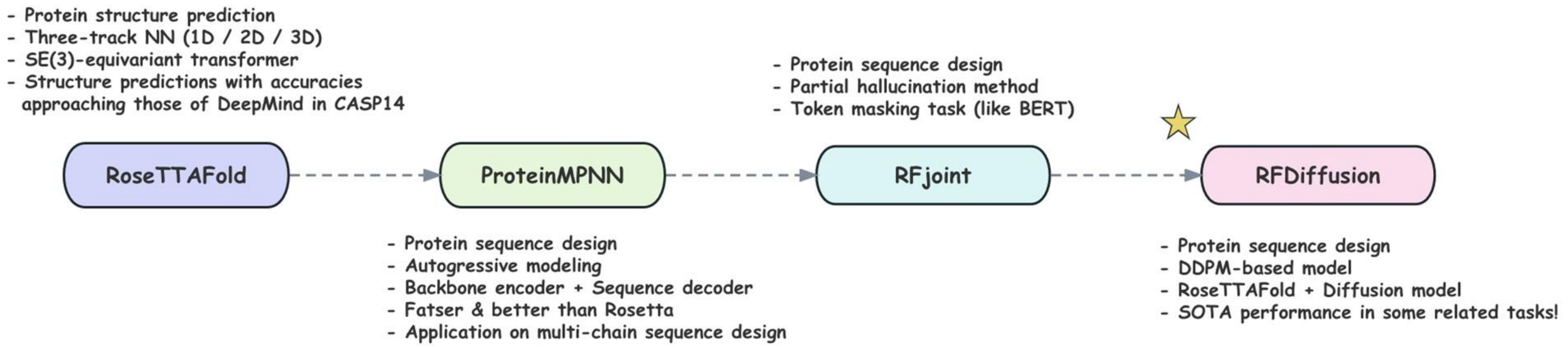


- 其中，Diffusion正是一个良好的赋能手段。2022-2023年也是Diffusion在计算生物学领域开始爆发的两年
- 主要集中于：**小分子生成**、**分子对接**、**蛋白质生成/蛋白质序列设计**等

Tasks	Applications	Frame	Representative Methods
Molecule Modeling	Molecule Conformation Generation	SMLD	MDM [Huang *et al.*, 2022]
		DDPM	GeoDiff [Xu *et al.*, 2022], EDMs [Hoogeboom *et al.*, 2022], EEGSDE [Bao *et al.*, 2022], DiGress [Vignac *et al.*, 2022]
		SGM	Torsional Diffusion [Jing *et al.*, 2022], MOOD [Lee *et al.*, 2022], GDSS [Jo *et al.*, 2022], DGSM [Luo *et al.*, 2021a], DiffBridges [Wu *et al.*, 2022b]
	Molecular Docking	DDPM	FragDiff [Anonymous, 2023c], DiffLink [Igashov *et al.*, 2022], TargetDiff [Anonymous, 2023a], DiffBP [Lin *et al.*, 2022]
		SGM	DiffDock [Corso *et al.*, 2022]
Protein Modeling	Protein Generation	DDPM	SMCDiff [Trippe *et al.*, 2022], SiamDiff [Zhang *et al.*, 2022], DiffFold [Wu *et al.*, 2022a], ProSSDG [Anand and Achim, 2022], DiffAntigen [Luo *et al.*, 2022a], RFdiffusion [Watson *et al.*, 2022]
		SGM	ProteinSGM [Luo *et al.*, 2022a]
	Protein-ligand Complex Structure Prediction	DDPM	DiffEE [Nakata *et al.*, 2022]
		SGM	NeuralPLexer [Qiao *et al.*, 2022]

Fan, Wenqi, et al. *arXiv preprint* 2023.

RoseTTAFold发展演变历程



- 在一众扩散（Diffusion）模型中，美国华盛顿大学 David Baker 团队的RFdiffusion十分耀眼
- 从 RoseTTAFold (RF) 的**序列预测结构**，再到 ProteinMPNN 与 RFjoint 的**结构预测序列**，再到 RFdiffusion，将以上两种功能有机结合在一起，最终实现了利用 diffusion 同时生成结构和序列的能力
  - 这个能力来自于**AF2/RF强大的 Language Model 框架**以及 **Diffusion** 强大的生成能力

RFdiffusion

上海科技大学 ShanghaiTech University

Article

De novo design of protein structure and function with RFdiffusion

https://doi.org/10.1038/s41586-023-06415-8

Received: 14 December 2022

Accepted: 7 July 2023

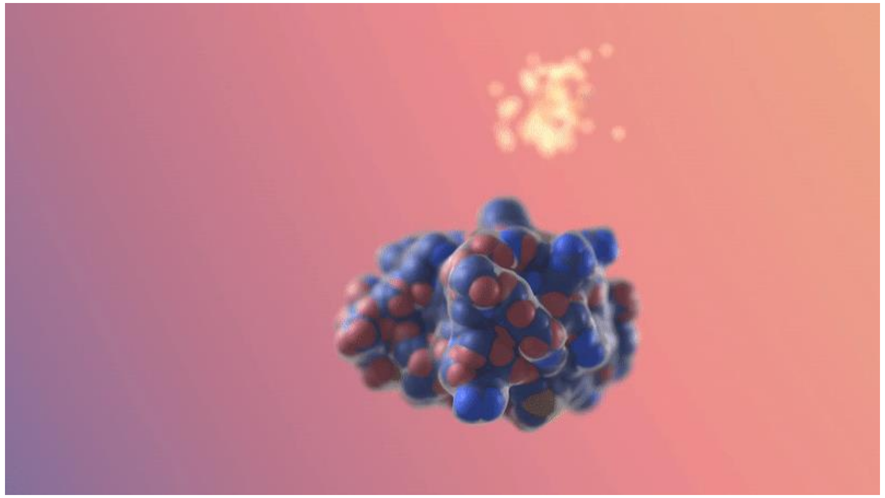
Published online: 11 July 2023

Open access

Joseph L. Watson[1,2,15], David Juergens[1,2,3,15], Nathaniel R. Bennett[1,2,3,15], Brian L. Trippe[2,4,5,15], Jason Yim[2,6,15], Helen E. Eisenach[1,2,15], Woody Ahern[1,2,7,15], Andrew J. Borst[1,2], Robert J. Ragotte[1,2], Lukas F. Milles[1,2], Basile I. M. Wicky[1,2], Nikita Hanikel[1,2], Samuel J. Pellock[1,2], Alexis Courbet[1,2,8], William Sheffler[1,2], Jue Wang[1,2], Preetham Venkatesh[1,2,9], Isaac Sappington[1,2,9], Susana Vázquez Torres[1,2,9], Anna Lauko[1,2,9], Valentin De Bortoli[8], Emile Mathieu[10], Sergey Ovchinnikov[11,12], Regina Barzilay[6], Tommi S. Jaakkola[6], Frank DiMaio[1,2], Minkyung Baek[13] & David Baker[1,2,14] ✉

立志成才 报国裕民

RFdiffusion



图为基于给定蛋白结合域的
binder蛋白随机生成

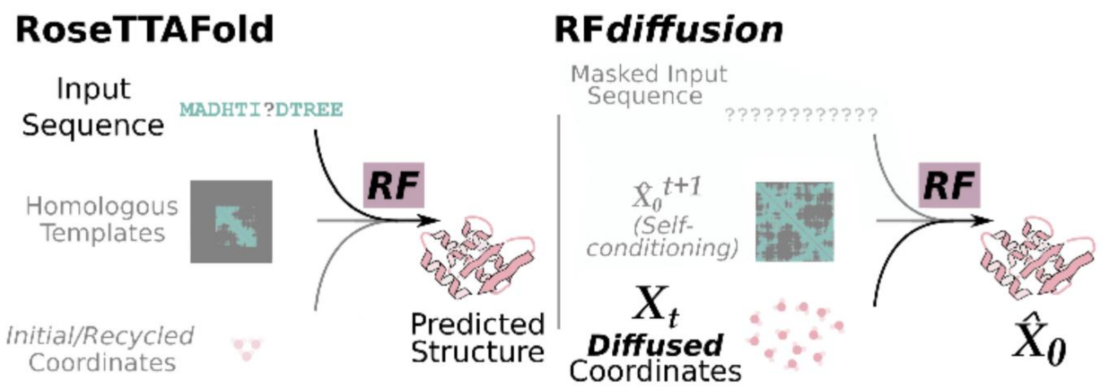
团队排名	Group Name	Country	University or Company
1	Yang-Server Yang	China	Shandong University，Nankai University, etc
2	UM-TBM	USA	University of Michigan, Michigan State University
3	PEZYFoldings	Japan	Infinite Curation Inc.
4	Dfolding Dfolding-server	Korea	Hanyang University, Korea Institute for Advanced Study, etc
5	McGuffin	UK	University of Reading
6	MULTICOM MULTICOM_human MULTICOM_refine MULTICOM_egnn MULTICOM_deep MULTICOM_qa	USA	University of Missouri
7	Kiharalab	USA, Japan	Purdue University, Tokyo Institute of Technology, etc
8	Manifold-E Manifold	China	DP technology
9	BAKER	USA	University of Washington
10	ShanghaiTech	China	ShanghaiTech University

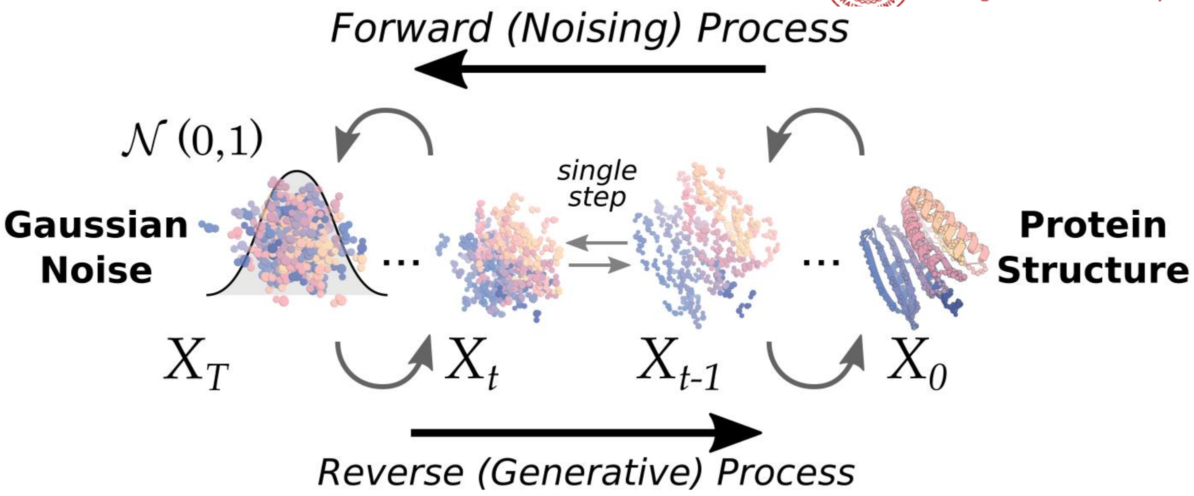
CASP15主赛道前10名

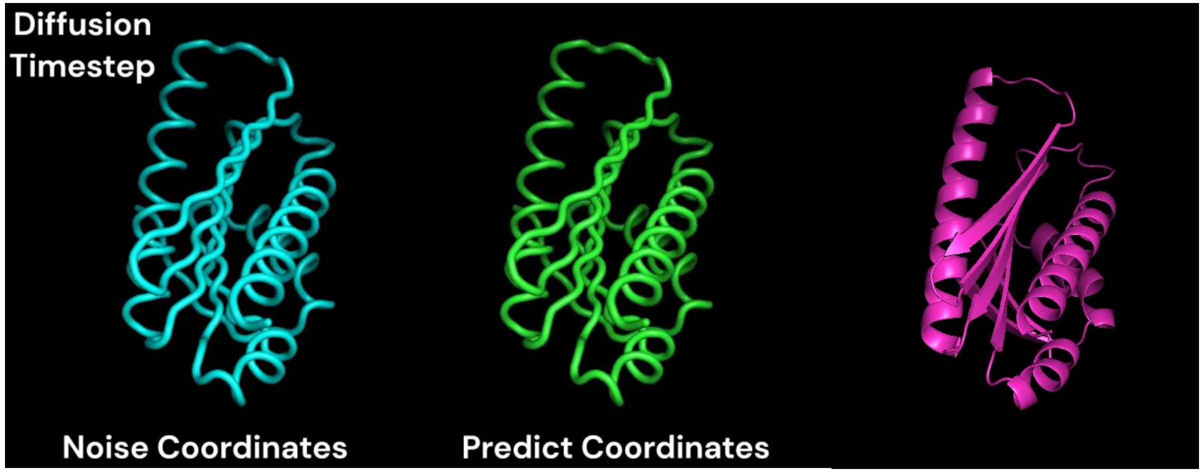
David Baker 团队的 RFdiffusion 模型在国际蛋白预测竞赛CASP15的主赛道中取得全球第9名次。

一个生成模型能在预测竞赛中取得如此斐然成绩，令人印象深刻。

Main idea of RFdiffusion



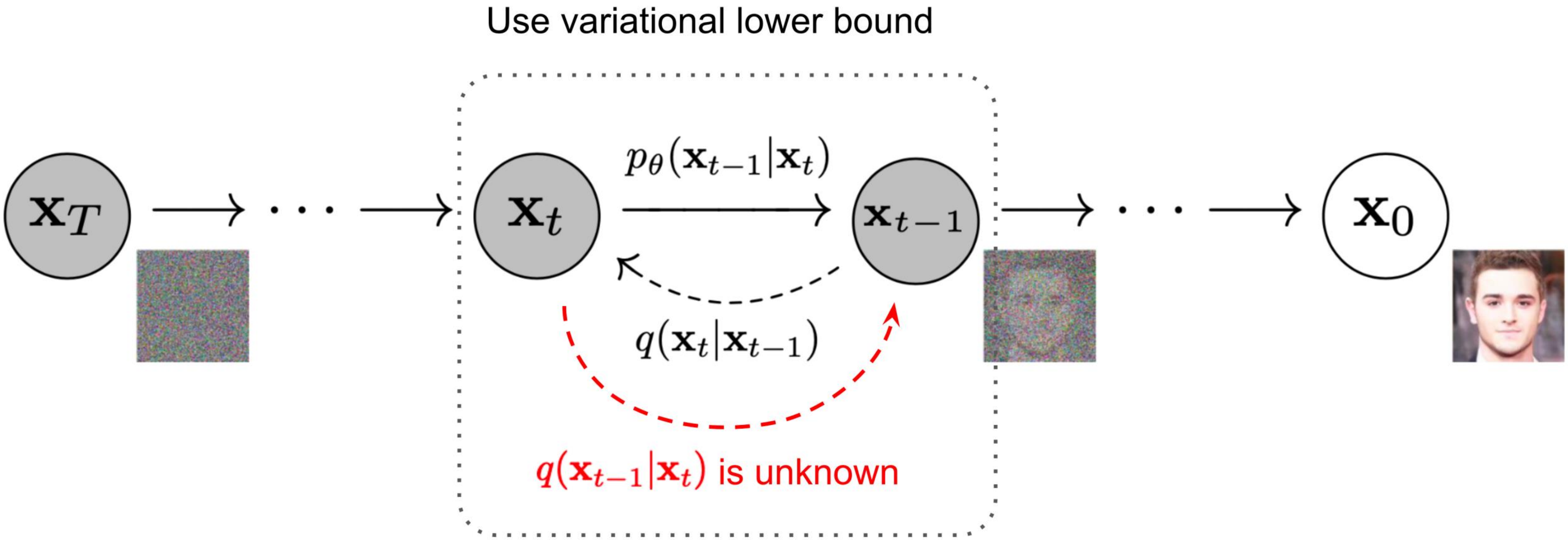




1. RFdiffusion 将每一个扩散时间步称为一个 **Sampler**（共50步）
2. **Sampler** 由 **RoseTTAFold** 和一个 **Denoiser** 构成
3. **RoseTTAFold** 由一个 **LM** 和一个 **3D等变结构模块** 组成（**类似于AF2**）
4. 在生成过程中，随着时间步降低，Denoiser 输出的**噪声坐标**会越来越接近**生成坐标**

RFdiffusion Self-conditioning Forward演示。
反向则是随机结构生成的过程。

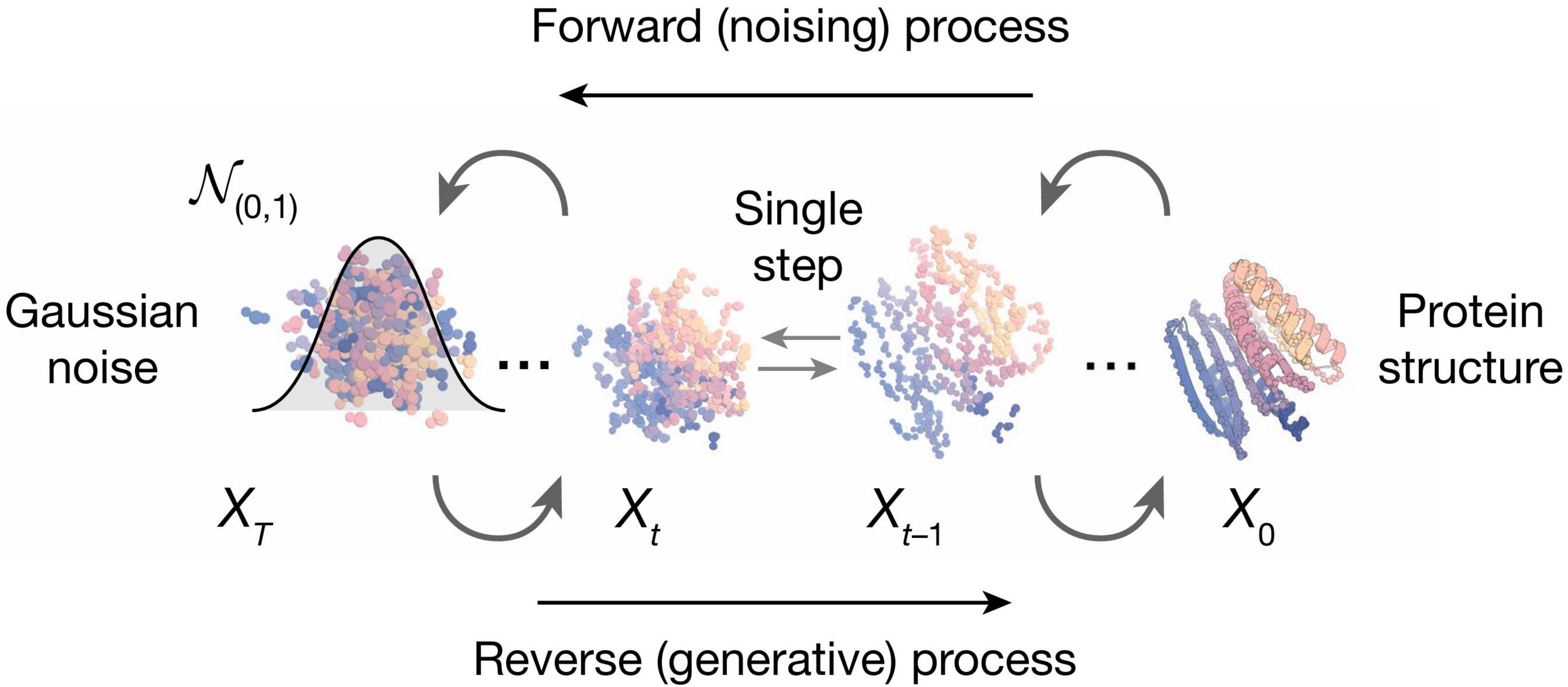
Denoising Diffusion Probabilistic Models (DDPMs)



Ho, Jonathan, Ajay Jain, and Pieter Abbeel. NeurIPS 2020.

DDPMs for protein design

Diffusion model

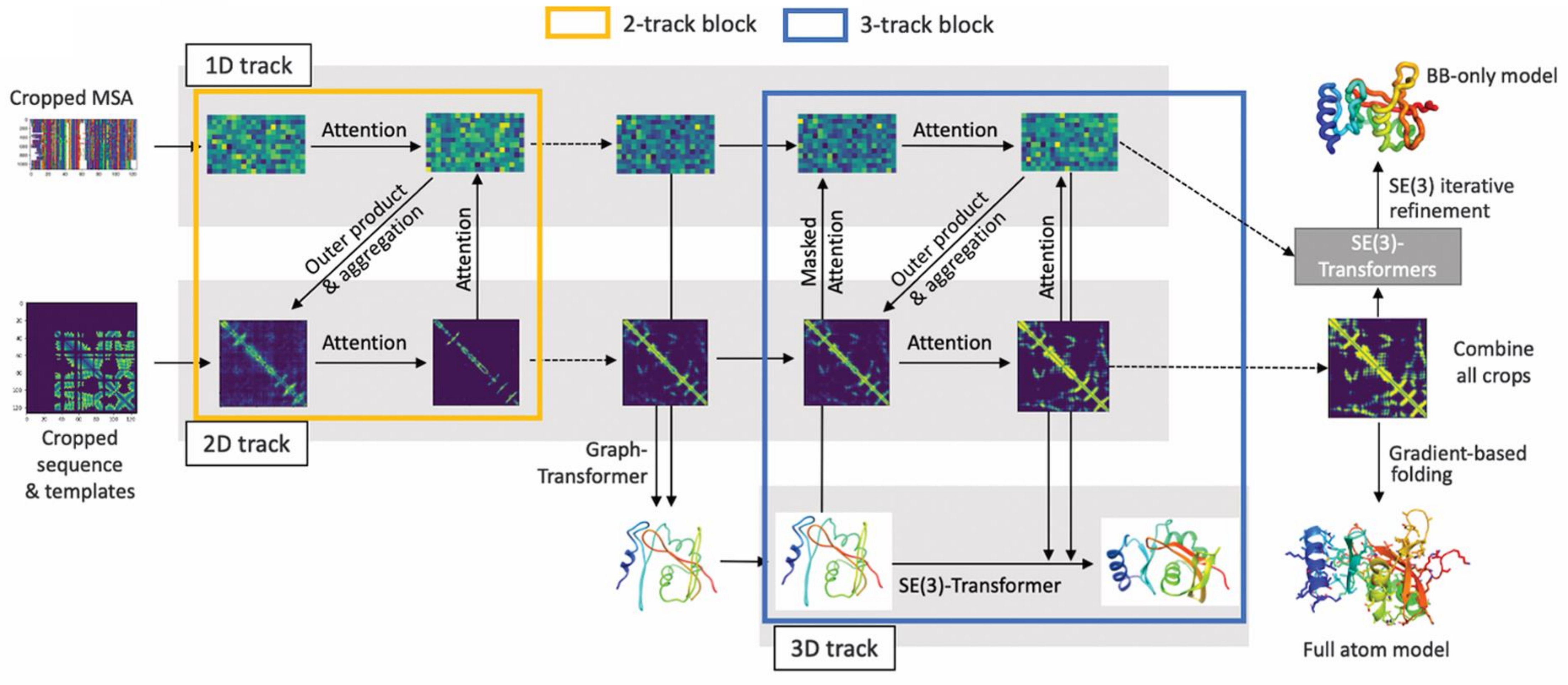


Watson, J.L., Juergens, D., Bennett, N.R. et al. Nature 2023.

Why choose DDPMs for protein design?

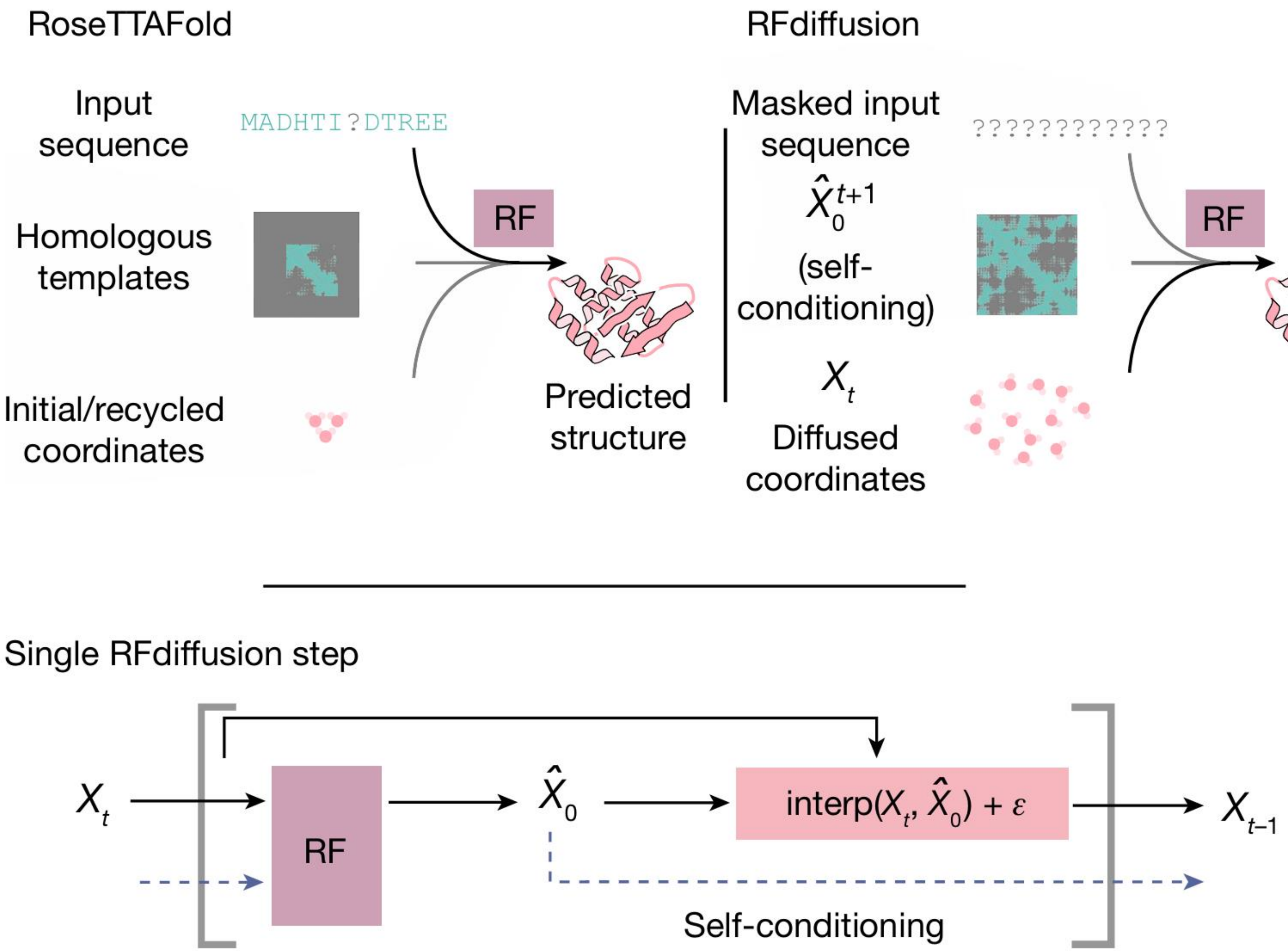
- DDPMs have several properties well-suited to protein design
  - Highly diverse outputs
  - Conditioning generation at each step
  - Explicitly 3D structure modeling via SE(3)-equivariant DDPMs
- Previous DDPMs-related methods had limited success
  - Limited ability of the denoising networks to generate realistic protein backbones
  - No experimental validation
- Need a powerful protein structure prediction framework that can act as a denoiser

RoseTTAFold



Baek et al., Science 373, 871–876 (2021)

RoseTTAFold vs. RFDiffusion



Watson, J.L., Juergens, D., Bennett, N.R. et al. Nature 2023.

References

- Madani et al. “Large language models generate functional protein sequences across diverse families”, Nature Biotechnology, 2022 (**ProGen**).
- Nijkamp et al. “ProGen2: Exploring the boundaries of protein language models”, arXiv, 2022 (**ProGen2**)
- Ho et al. “Denoising diffusion probabilistic models”, NeurIPS, 2020 (**DDPM**).
- J. L. Watson et al. “De novo design of protein structure and function with RFdiffusion”, Nature, Vol 620, pp. 1089-1100, 2023 (**RFdiffusion**).